

VICTORIA PARK ACADEMY SHENZHEN

#

IGCSE COMPUTER SCIENCE (0478) — YEAR

10 SEMESTER 2

#

WEEKS 10–18: GAP ANALYSIS • TARGETED

REVISION • MOCK SERIES 2

#

FINAL EXAM PREPARATION

#

Curriculum Document — Complete with

Lesson Plans, Worksheets,

#

Mock Exam Papers, Mark Schemes &

Answer Keys

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CLASS PROFILE:

- Year 10 students (age 15–16) preparing for CIE IGCSE Computer Science (0478)
- EAL learners — bilingual support (English / Chinese / Pinyin) embedded
- Mixed ability: Emerging / Developing / Secure
- Lesson duration: 40 minutes
- Two lessons per week
- Class size: typically 15–24 students

ASSESSMENT JOURNEY TO DATE:

Mock 1 → Y10 Semester 1 Weeks 17–18 (Paper 1 & Paper 2)

Mock 2 → Y10 Semester 2 Week 9 (Paper 1 — 75 marks)

Mock 3 → Y10 Semester 2 Week 10 (Paper 2 — 75 marks)

Mock 4 → Y10 Semester 2 Week 15 (Paper 1 — 75 marks) ← THIS DOCUMENT

Mock 5 → Y10 Semester 2 Week 16 (Paper 2 — 75 marks) ← THIS DOCUMENT

STRUCTURE OF THIS DOCUMENT:

Unit 4 (Weeks 10–14): Gap Analysis & Targeted Revision

Unit 5 (Weeks 15–16): Mock Series 2 (Mocks 4 & 5)

Unit 6 (Weeks 17–18): Final Preparation & Exam Readiness

For each week:

- 2 × 40-minute lesson plans with full structure
- 2 × student worksheets (with answer keys)
- EAL support & differentiation (Emerging/Developing/Secure)
- Full mock papers with complete mark schemes (Weeks 15–16)

GRADE BOUNDARIES (CIE IGCSE 0478 — indicative):

A*: 68–75 (Paper 1 or 2)

A: 60–67

B: 52–59

C: 44–51

D: 36–43

E: 28–35

F: 20–27

G: 12–19

U: 0–11

UNIT 4 OVERVIEW

Gap Analysis & Targeted Revision

Unit 4 spans Weeks 10–14 and represents the most personalised phase of the

entire Year 10 programme. Every activity is data-driven: mock results from Mocks 1, 2 and 3 are combined to produce individual gap analyses, which then drive targeted revision groups, differentiated worksheets, and focused intervention. The goal is not to re-teach the entire syllabus, but to maximise each student's marks by addressing their specific weaknesses efficiently.

Week 10: Mock 3 Review & Gap Analysis Preparation

Week 11: Individual Gap Analysis & Targeted Revision Groups

Week 12: Targeted Revision — Paper 1 Weak Areas

Week 13: Targeted Revision — Paper 2 Weak Areas

Week 14: Mixed Practice & Consolidation

WEEK 10: MOCK 3 REVIEW & GAP ANALYSIS PREPARATION

OVERVIEW:

Week 10 is a transition week. Mock 3 (Paper 2) was completed in Week 9 or

early Week 10. These two lessons review that mock, celebrate successes, and begin the process of compiling all mock data to prepare for individual gap analysis in Week 11. Students reflect on their Paper 2 performance and start to think metacognitively about which topics they personally find most challenging.

LESSON 10.1 — MOCK 3 (PAPER 2) REVIEW AND REFLECTION

Topic: Mock Exam Analysis & Error Review

Duration: 40 min | Lesson Type: Review & Reflection | Syllabus Ref: All Paper 2

Learning Objective:

By the end of this lesson, students will be able to identify their own errors in Mock 3 (Paper 2), classify each error type (knowledge gap, misreading, calculation error, time pressure), and set one specific target.

Key Vocabulary:

- gap analysis (差距分析 chājù fēnxī) — identifying what you don't know
- error classification (错误分类 cuòwù fēnlèi) — sorting mistakes by type
- metacognition (元认知 yuán rènzhī) — thinking about your own thinking
- misconception (误解 wùjiě) — a wrong understanding of a concept
- target (目标 mùbiāo) — a specific goal to achieve

Lesson Structure (40 Minutes):

0–3 min | DO NOW

Students enter and receive their marked Mock 3 Paper 2. On their

"Error Analysis Sheet" (provided as Worksheet 10A), they immediately write down:

1. Their total mark out of 75
2. The one question they found hardest
3. One thing they think they did well

Teacher circulates and offers brief individual encouragement.

3–12 min | DIRECT INSTRUCTION

Teacher presents the class performance summary (prepared in advance):

- Class average, highest, lowest, and range
- Top 5 most-commonly-missed questions (displayed on board)
- Question-by-question class average (bar chart projected)

Teacher models the error classification process using the class's most-missed question:

- Was it a KNOWLEDGE gap? (didn't know the concept)
- Was it a READING error? (misread the question)
- Was it a CALCULATION error? (arithmetic slip)
- Was it TIME pressure? (ran out of time)
- Was it a CARELESS error? (knew it but made a silly mistake)

Show students how to colour-code their paper: pink = knowledge gap, yellow = reading error, blue = calculation, green = time pressure.

12–25 min | GUIDED PRACTICE

Students work in pairs. They compare their Mock 3 papers and discuss:

- Which questions did we both get wrong? Why?
- Which questions did one get right and the other wrong? Can the stronger student explain their thinking?

Teacher circulates and listens for common themes. Pairs complete the first half of their Error Analysis Sheet together.

25–35 min | INDEPENDENT PRACTICE

Students complete their personal Error Analysis Sheet independently:

- List their 5 weakest questions with topic names
- Classify each error (pink/yellow/blue/green)
- Write one specific target for each error type
- Calculate their own percentage and grade

Teacher conferences with 3–4 students individually (30–60 sec each).

35–40 min | PLENARY + HOMEWORK

"One thing I will improve before the final exams is..." — students share with a partner, then 3 volunteers share with the class.

HOMEWORK: Bring to next lesson — a list of 3 topics you think are your weakest areas (before seeing the official gap analysis).

EAL Support:

- Error Analysis Sheet includes Chinese translations for all error types
- Sentence starters provided in English and Chinese for reflection
- Teacher checks understanding with EAL students individually during independent practice
- Visual colour-coding reduces language barrier

Differentiation:

Emerging (below 35/75):

- Teacher provides pre-filled error categories (student just matches)
- Focus on knowledge gaps only (ignore time pressure for now)
- One-to-one check-in with teacher during lesson

Developing (35–55/75):

- Full Error Analysis Sheet with scaffolding prompts
- Encouraged to focus on 2–3 priority topics, not everything

Secure (above 55/75):

- Complete full analysis independently
- Extension: analyse the mark scheme to identify exactly what wording would have earned lost marks
- Mentor a Developing student during pair work

Resources Needed:

- Marked Mock 3 Paper 2 papers (returned to students)
- Projected class performance summary (bar chart)
- Worksheet 10A — Error Analysis Sheet (one per student)
- Highlighters in 4 colours (pink, yellow, blue, green)
- Sentence starters on display (EN/CN/pinyin)

LESSON 10.2 — COMPILING THE GAP ANALYSIS

Topic: Data-Driven Self-Assessment & Topic Mapping

Duration: 40 min | Lesson Type: Workshop | Syllabus Ref: All

Learning Objective:

By the end of this lesson, students will be able to combine their mock results across all three assessments, produce a personal topic-level performance map, and identify their top 3 priority topics for revision.

Key Vocabulary:

- compile (汇编 huìbiān) — to bring together from different sources
- performance map (表现图 biǎoxiàn tú) — a visual showing how well you did
- priority topic (优先主题 yōuxiān zhǔtí) — the most important thing to revise

- aggregate (总计 zǒngjì) — total combined score
- diagnostic (诊断 zhěnduàn) — finding out what is wrong

Lesson Structure (40 Minutes):

0–3 min | DO NOW

Students receive their "Personal Performance Tracker" (Worksheet 10B).

They fill in their marks for:

Mock 1 Paper 1: ____/75 Mock 1 Paper 2: ____/75

Mock 2 Paper 1: ____/75 Mock 3 Paper 2: ____/75

Teacher has pre-filled this for students who may not remember.

3–12 min | DIRECT INSTRUCTION

Teacher explains the tracking system:

- Each Paper 1 question tests specific topics (display topic-question mapping on board)
- Each Paper 2 question tests specific topics (display mapping)
- We can work backwards from wrong answers to identify weak TOPICS

Teacher models with a worked example on the board:

"If I got Q3 wrong on Mock 2 and Q7 wrong on Mock 4, and both test 'file size calculations', then that's a priority topic — not just two random wrong answers."

Show the class-level "Topic Difficulty Chart" (teacher-prepared in advance from all mock data):

Rank topics by % of class marks lost:

4. Pseudocode with loops (78% of class lost marks)
5. SQL JOIN queries (72% of class lost marks)
6. File size calculations (68% of class lost marks)
7. Boolean logic circuit design (65% of class lost marks)
8. CPU architecture/fetch-execute (60% of class lost marks)

[These are examples — teacher should replace with actual class data]

12–25 min | GUIDED PRACTICE

Students work with the topic-question mapping to identify their own weak topics. Using the projected mapping, they:

- Go through each mock and note which topics their wrong answers belong to
- Tally marks next to each topic on their tracker
- Circle any topic with 2+ wrong answers across all mocks

Teacher circulates with the official gap analysis printout — each student receives their pre-prepared "Official Gap Analysis" slip (teacher has prepared these from markbooks). Students compare their self-analysis with

the official analysis and discuss any differences with teacher.

25–35 min | INDEPENDENT PRACTICE

Students complete their Personal Performance Tracker:

- Top 3 weak topics (personal)
- Predicted grade based on average of all mocks
- Target grade
- Specific actions needed for each of top 3 topics

Students write their top 3 topics on a sticky note and post it on the class "Revision Target Wall" (organized by topic area: Data Rep, Hardware, Programming, Databases, etc.).

35–40 min | PLENARY + HOMEWORK

Teacher reveals the 3 most popular weak topics from the sticky notes.

"These are the class priorities — we will cover these together."

Explain that next lesson, students will be grouped by weak area for targeted revision.

HOMEWORK: Complete any unfinished tracker sections. Research one fact about each of your top 3 weak topics (use textbook or notes).

EAL Support:

- Official Gap Analysis slip has Chinese topic translations
- Topic-question mapping poster on wall (permanent reference)
- Teacher assistant or bilingual peer supports EAL students during self-analysis
- Visual tracker uses colours and icons, not just text

Differentiation:

Emerging:

- Teacher provides pre-completed topic tally (student just confirms)
- Top 3 topics identified by teacher, not student
- Simplified tracker with fewer topic categories

Developing:

- Full tracker with scaffolding (sentence starters for action plans)
- Prompt questions to guide self-reflection

Secure:

- Full tracker completed independently
- Extension: calculate their own percentage per topic
- Support classmates by explaining the mapping process

Resources Needed:

- Personal Performance Tracker (Worksheet 10B) — one per student
- Official Gap Analysis slips (teacher-prepared, one per student)
- Topic-question mapping projection/posters
- Class Topic Difficulty Chart projection
- Sticky notes in different colours
- Revision Target Wall display board

--- WORKSHEET 10A ---

WORKSHEET 10A: MOCK 3 ERROR ANALYSIS SHEET

Name: _____ Class: _____ Date: _____

PART A: MY RESULTS

1. My total mark: _____ / 75

2. My percentage: _____% (mark ÷ 75 × 100)

3. My approximate grade: _____ (use grade boundaries)

9. Compared to Mock 2 (Paper 1), I have: ☐ improved ☐ stayed the same ☐ dropped

PART B: MY FIVE WEAKEST QUESTIONS

For each question you got wrong or partially wrong, fill in the table:

Q	Topic (in English and 中文)	My mark	Max	Error type	Why I got it wrong
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Error Type Key (错误类型):

K = Knowledge gap (知识差距 zhīshì chājù) — I didn't know this

R = Reading error (阅读错误 yuèdú cuòwù) — I misread the question

C = Calculation error (计算错误 jìsuàn cuòwù) — arithmetic slip

T = Time pressure (时间压力 shíjiān yālì) — I ran out of time

X = Careless mistake (粗心错误 cūxīn cuòwù) — I knew it but made a silly error

PART C: ERROR SUMMARY

Count your errors:

Knowledge gaps: _____ Reading errors: _____ Calculation errors: _____

Time pressure: _____ Careless mistakes: _____

PART D: MY TARGETS

Write one specific target for your biggest error type:

Write one specific target for a topic you need to revise:

PART E: ONE THING I DID WELL

--- ANSWER KEY: WORKSHEET 10A ---

WORKSHEET 10A ANSWER KEY & TEACHER GUIDANCE

This is a reflective worksheet — there are no "right" answers. However, teachers should use this guidance to ensure students complete it meaningfully.

MARKING CRITERIA (formative — completion-based, 10 marks):

- Part A completed correctly (2 marks)
 - Percentage calculation correct (1)
 - Grade identified correctly from boundaries (1)
- Part B: at least 3 questions analysed with all columns filled (3 marks)
 - Each row: topic named (0.5), error type classified (0.5)
 - 3 rows × 2 components = 3 marks (or 5 rows for more detail)
- Part C: error counts match Part B (1 mark)
- Part D: both targets are SPECIFIC and ACTIONABLE (2 marks)
 - Good example: "I will practise 5 file size calculation questions"
 - Weak example: "I will try harder" (not specific enough)
- Part E: positive reflection written (1 mark)
- Overall presentation and effort (1 mark)

TEACHER NOTES:

- Review all sheets before Lesson 10.2 — flag any student whose targets are not specific enough and provide written feedback
- Look for patterns across the class — if 50%+ have "K" for pseudocode, that topic needs whole-class revision
- Use Part B data to help complete the gap analysis in Lesson 10.2
- This sheet is designed to build metacognitive skills — encourage honesty in error classification

--- WORKSHEET 10B ---

WORKSHEET 10B: PERSONAL PERFORMANCE TRACKER

Name: _____ Target Grade: _____ Predicted Grade: _____

PART A: MY MOCK RESULTS JOURNEY

Mock	Paper	Topics Tested	My Mark /75	%	Grade	Topics I Got Wrong
M1	P1	Hardware, Data Rep, Networks, Security				
M1	P2	Algorithms, Programming, Databases, Boolean				

M2	P1	Full Paper 1 syllabus				
M3	P2	Full Paper 2 syllabus				

PART B: MY TOPIC-LEVEL ANALYSIS

For each topic, count how many questions you got **WRONG** across all mocks:

Paper 1 Topics:

Topic (EN)	Topic (中文)	Wrong (tally)	Priority?
Number systems (binary/hex)	数制系统		
Binary arithmetic	二进制运算		
Text, sound, image representation	文本声音图像表示		
File size calculations	文件大小计算		
Compression (lossy/lossless)	压缩		
Hardware (input/output/storage)	硬件		
Logic gates and circuits	逻辑门电路		
CPU architecture & F-E cycle	CPU 架构		
Memory (RAM/ROM/Cache)	内存		
Secondary storage	辅助存储		
Operating systems & utility software	操作系统		
Networks (LAN/WAN/topologies)	网络		
Network hardware & protocols	网络硬件协议		
Internet, WWW, IP addressing	互联网		
Security threats & prevention	安全威胁		
Emerging technologies	新兴技术		

Paper 2 Topics:

Topic (EN)	Topic (中文)	Wrong (tally)	Priority?
Algorithm design & flowcharts	算法设计		
Pseudocode	伪代码选择		

(sequence/selection)			
Pseudocode (iteration/loops)	伪代码循环		
Pseudocode (arrays/file handling)	伪代码数组文件		
Trace tables	追踪表		
Python programming basics	Python 基础		
Python (selection/iteration)	Python 选择循环		
Python (functions/arrays)	Python 函数数组		
Program design & testing	程序设计测试		
SQL queries (SELECT/WHERE)	SQL 查询		
SQL (JOIN/aggregate)	SQL 连接聚合		
Database design & normalisation	数据库设计		
Boolean algebra & simplification	布尔代数		
Logic circuit design	逻辑电路设计		
15-mark scenario question	15 分情景题		

PART C: MY TOP 3 PRIORITY TOPICS

10. _____ (wrong ____ times across all mocks)

Action plan: _____

11. _____ (wrong ____ times across all mocks)

Action plan: _____

12. _____ (wrong ____ times across all mocks)

Action plan: _____

PART D: MY PROGRESS GOAL

My average mock mark: _____ / 75 (____%)

My target for Mock 4/5: _____ / 75 (____%)

To reach my target, I need to improve by: _____ marks

The topic that will give me the MOST extra marks is: _____

--- ANSWER KEY: WORKSHEET 10B ---

WORKSHEET 10B ANSWER KEY & TEACHER GUIDANCE

This is a personalised reflective tracker — answers vary per student.

This key provides the marking framework and correct calculations.

PART A (4 marks — completion):

- All 4 mock rows completed with marks filled in (2 marks)
- All percentages calculated correctly (1 mark)
 - Formula: $(\text{mark} \div 75) \times 100$, rounded to 1 decimal place
- Grades identified correctly from grade boundaries (1 mark)

PART B (4 marks — analysis):

- Paper 1 topic tally completed with realistic data (2 marks)
- Paper 2 topic tally completed with realistic data (2 marks)
- "Priority" circled for any topic with 2+ wrong answers

PART C (4 marks — targets):

- 3 priority topics identified with evidence from Part B (2 marks)
- Each action plan is SPECIFIC and ACHIEVABLE (2 marks)
 - Good: "Practise 5 SQL JOIN questions from revision booklet"
 - Weak: "Try harder at SQL" (deduct 0.5 per weak plan)

PART D (3 marks — goal-setting):

- Average calculated correctly (1 mark)
 - Formula: $(M1P1 + M1P2 + M2 + M3) \div 4$ [or $\div 3$ if one absent]
- Target set realistically (above current average) (1 mark)
- Improvement topic selected logically (1 mark)

TOTAL: 15 marks (formative)

TEACHER NOTES:

- Pre-fill Part A for students who were absent for any mock
- Check that Part C action plans are specific before signing off
- Use completed trackers to form targeted revision groups for Week 11
- Photocopy completed sheets for teacher records before students take them home
- Students should bring this tracker to EVERY revision lesson — it is their personalised revision roadmap

SAMPLE CALCULATIONS (for teacher reference):

Student with M1P1=32, M1P2=38, M2=35, M3=42:

Average = $(32 + 38 + 35 + 42) \div 4 = 147 \div 4 = 36.75 \approx 36.8/75 = 49.1\%$

Target for M4/5: 45/75 (60%) — improvement of 8.3 marks

WEEK 11: INDIVIDUAL GAP ANALYSIS

OVERVIEW:

This is the most personalised week of the entire course. Every student receives their individual gap analysis compiled from all three mocks.

Students are grouped by weak area and begin targeted revision in rotating small groups. The teacher moves between groups, providing intensive support. Differentiated worksheets ensure each student works on material precisely targeted at their needs.

LESSON 11.1 — DATA-DRIVEN REVISION & GROUPING

Topic: Individual Gap Analysis & Revision Group Formation

Duration: 40 min | Lesson Type: Data Workshop | Syllabus Ref: All

Learning Objective:

By the end of this lesson, students will be able to interpret their individual gap analysis data, identify their personal top 3 weak topics, and join a targeted revision group appropriate to their needs.

Key Vocabulary:

- data-driven (数据驱动 shùjù qūdòng) — using data to make decisions
- revision group (复习小组 fùxí xiǎozǔ) — a small group studying together
- rotate (轮换 lúnhuàn) — to move from one group to another
- peer support (同伴支持 tóngbàn zhīchí) — helping each other learn
- cohort (同组同学 tóng zǔ tóngxué) — students in your group

Lesson Structure (40 Minutes):

0–3 min | DO NOW

Students enter and find their "Individual Gap Analysis Card" on their desk.

Card shows:

- Student name and all mock marks
- Ranked list of weak topics (1 = weakest)
- Suggested revision group (A, B, or C)

Students read their card silently and circle their top 3 topics.

3–12 min | DIRECT INSTRUCTION

Teacher explains the gap analysis card:

"This card shows which topics you lost the most marks on across ALL three mocks. It's personalised — your card is different from your neighbour's."

Teacher models interpretation using an anonymous example card on the board:

- "This student lost marks on: SQL JOINS (3 times), Boolean circuits (2 times), and file size calculations (2 times). So their suggested group is Group C (Databases/SQL/Boolean)."

Teacher introduces the class "Topic Difficulty Chart":

- Bar chart showing whole-class performance by topic
- Class votes: "Raise your hand if you found SQL hard" → visual count

- Top 3 class-wide weak topics identified for whole-class revision later

REVISION GROUPS EXPLAINED:

Group A — Data Representation & Hardware (数据表示与硬件)

Topics: binary, hex, file sizes, compression, CPU, memory, storage

Group B — Programming & Algorithms (编程与算法)

Topics: pseudocode, trace tables, Python, program design, testing

Group C — Databases, SQL & Boolean (数据库、SQL 与布尔)

Topics: SQL queries, database design, Boolean algebra, logic circuits

12–25 min | GUIDED PRACTICE

Students move to their assigned revision group areas (3 areas in room).

Each area has:

- Colour-coded group sign (A=red, B=blue, C=green)
- Topic checklist poster
- Starter activity on table

Group A starter: Convert 10110101 binary to denary and hex

Group B starter: Trace this pseudocode loop for 3 iterations

Group C starter: Write an SQL query to find all customers from 'Shenzhen'

Students work on starter while teacher circulates. Teacher spends 3–4 minutes with each group checking understanding and setting expectations.

25–35 min | INDEPENDENT PRACTICE

Students begin their Group Revision Pack (Worksheet 11A):

- Section 1: Knowledge check (5 quick questions on their weak topics)
- Section 2: Worked example with gaps to fill
- Section 3: Independent practice questions

Teacher circulates intensively — spends 5 min in each group area:

- Group A: supports file size calculations and binary conversions
- Group B: helps with pseudocode syntax and trace table technique
- Group C: guides SQL query structure and Boolean simplification

35–40 min | PLENARY + HOMEWORK

Each group reports: "One thing we practised today and one thing we're still unsure about" (30 seconds per group).

Teacher announces that next lesson, groups will rotate to a second topic OR continue with their first topic (depending on need).

HOMEWORK: Complete Section 3 of the Group Revision Pack. Bring questions for next lesson.

EAL Support:

- Gap Analysis Card has all topics in English, Chinese, and pinyin
- Group area signs include visual icons for each topic
- Bilingual vocabulary list provided at each group station
- Group starters use visual/scaffolded formats where possible
- Teacher or bilingual peer check EAL students understand their gap analysis

Differentiation:

Emerging:

- Teacher identifies their group and escorts them there
- Simplified revision pack (fewer questions, more scaffolding)
- Pre-completed worked example (student just follows pattern)
- Pair with a secure buddy in same group

Developing:

- Full revision pack with structured support
- Sentence starters for explaining reasoning
- Checklist to self-monitor understanding

Secure:

- Full revision pack + extension questions
- Extension: create a revision resource (flashcard/poster) for their group
- Mentor role: support an emerging student after completing own work

Resources Needed:

- Individual Gap Analysis Cards (teacher-prepared, one per student)
- Class Topic Difficulty Chart (projected)
- Worksheet 11A — Group Revision Packs (3 versions: A, B, C)
- Group area signs (A, B, C) with topic lists
- Starter activities for each group
- Bilingual vocabulary lists for each group area

LESSON 11.2 — TARGETED REVISION: THREE GROUPS RUNNING

Topic: Differentiated Targeted Revision — Rotating Group Support

Duration: 40 min | Lesson Type: Rotating Workshop | Syllabus Ref: All

Learning Objective:

By the end of this lesson, all students will have received intensive teacher-supported revision on their weakest topic area and produced a personal revision plan for the final exam period.

Key Vocabulary:

- revision plan (复习计划 fùxí jìhuà) — a schedule for what to revise when
- intensive support (强化支持 qiáng huà zhī chí) — focused help
- rotate (轮换 lún huàn) — move to the next station
- checkpoint (检查点 jiǎn chá diǎn) — a point to check understanding

- autonomous (自主的 zìzhǔ de) — working independently

Lesson Structure (40 Minutes):

0–3 min | DO NOW

Students sit in their revision groups from Lesson 11.1. On their "Personal Revision Plan" template (Worksheet 11B), they write:

- Today's date
- Days until final mock exams (countdown)
- Their revision group (A, B, or C)
- One question they want answered today

3–12 min | DIRECT INSTRUCTION — ROTATION 1

Teacher works intensively with ONE group (e.g., Group C — Databases/SQL):

- 3 mini-lessons in one: explains common errors, models correct approach, gives immediate practice
- Uses mini-whiteboards for instant assessment
- Addresses the specific misconceptions this group has shown in mocks

Other two groups work independently on their revision packs with answer keys available for self-checking. A "Help Card" system: if stuck, students check the 3 hint cards before asking the teacher.

12–25 min | DIRECT INSTRUCTION — ROTATION 2

Teacher moves to second group (e.g., Group B — Programming/Algorithms):

- Same intensive format: common errors → model → practice
- Focus on pseudocode syntax rules (CAPS, ← not =, ENDIF/ENDWHILE)
- Trace table technique: one step per box, watch for loop counters

Remaining group continues independent work with self-checking.

25–35 min | DIRECT INSTRUCTION — ROTATION 3

Teacher moves to third group (e.g., Group A — Data Rep/Hardware):

- Common errors: binary to hex conversion (group by 4 from RIGHT)
- File size formula: width × height × depth (for images), or sampling rate × duration × depth (for sound)
- CPU fetch-execute cycle: fetch → decode → execute → store

Final independent group self-checks and notes any remaining questions.

35–40 min | PLENARY + HOMEWORK

Students return to their regular seats. Complete the Personal Revision Plan:

- What I learned today (one thing from my group)
- What I still need to revise before the mock
- My revision schedule for the next 2 weeks

Collect revision plans for teacher review.

HOMEWORK: Follow the revision plan — minimum 20 minutes per night on identified weak topics. Complete any unfinished pack questions.

EAL Support:

- All hint/help cards bilingual (EN/CN)
- Teacher uses simplified English + visuals during group instruction
- Revision plan template has Chinese translations for all sections
- EAL students can write revision plan in Chinese first, then translate
- Visual timers shown so students know when teacher will rotate

Differentiation:

Emerging:

- Stay in same group for both rotations (no change needed)
- Teacher prioritises this group for first rotation (most alert time)
- Revision plan has pre-filled time slots (student just adds topic names)
- Daily check-in with teacher or LSA

Developing:

- Rotate as planned
- Revision plan has suggested topics for each day, student confirms
- Self-check answers using provided answer keys

Secure:

- Rotate as planned
- Complete revision plan independently with stretch targets
- During teacher's rotation to other groups, act as group mentor
- Extension: create a 5-question quiz on their weak topic for peers

Resources Needed:

- Worksheet 11A — Group Revision Packs (3 versions, differentiated)
- Worksheet 11B — Personal Revision Plan template
- Mini-whiteboards and pens for each group
- Help/hint cards (3 per group, bilingual)
- Answer keys for all revision pack questions
- Visual timer/stopwatch for teacher rotations
- Countdown calendar (days until mocks)

--- WORKSHEET 11A ---

WORKSHEET 11A: GROUP REVISION PACKS (3 Versions)

VERSION A: DATA REPRESENTATION & HARDWARE

For students whose gap analysis shows weakness in: binary/hex, file sizes, compression, CPU, memory, or storage.

SECTION 1: KNOWLEDGE CHECK (5 marks)

1. Convert 11001101 binary to denary. (1 mark)

2. Convert 9F hex to denary. (1 mark)

13. Name one type of lossy compression and one type of lossless compression.
(1 mark)

4. State the purpose of the ALU in the CPU. (1 mark)

5. Explain one difference between RAM and ROM. (1 mark)

SECTION 2: WORKED EXAMPLE — FILE SIZE CALCULATION (guided)

An image is 800×600 pixels with a colour depth of 24 bits.

Step 1: Write the formula: File size = width $\times$ height $\times$ colour depth

Step 2: Substitute: File size = $800 \times 600 \times$ _____

Step 3: Calculate: File size = _____ bits

Step 4: Convert to bytes: $\div 8 \rightarrow$ _____ bytes

Step 5: Convert to KB: $\div 1024 \rightarrow$ _____ KB

Now you try:

An image is 1024×768 pixels with a colour depth of 16 bits.

Calculate the file size in KB. Show all working. (3 marks)

SECTION 3: INDEPENDENT PRACTICE

Q1. Convert these binary numbers to hexadecimal: (3 marks)

a) 1110 0011 = _____

b) 1010 1100 1111 = _____

c) 0001 1000 1010 0101 = _____

Q2. A sound file is recorded at 44.1 kHz sampling rate, 16-bit sample depth, for 3 minutes. Calculate the file size in MB. Show all working. (4 marks)

Hint: remember sampling rate is in kHz — convert to Hz first!

Q3. Explain how run-length encoding (RLE) compresses an image. Use the following row of pixels in your answer: R R R R G B B B B W W (3 marks)

Q4. Draw and label a diagram of the CPU showing: ALU, CU, registers, and the address bus, data bus, and control bus. (4 marks)

Q5. Describe the purpose of cache memory and explain why a larger cache generally improves computer performance. (3 marks)

VERSION B: PROGRAMMING & ALGORITHMS

For students whose gap analysis shows weakness in: pseudocode, trace tables, Python programming, or program design.

SECTION 1: KNOWLEDGE CHECK (5 marks)

14. Write pseudocode to input a number and output "Positive" if it is > 0 , otherwise output "Not positive". (1 mark)

15. What value will x have after this loop finishes? (1 mark)

$x \leftarrow 0$

FOR $i \leftarrow 1$ TO 4

$x \leftarrow x + i$

ENDFOR

16. Name two types of test data, giving an example for each. (1 mark)

4. What does MOD do? Give an example. (1 mark)

17. State one advantage of using functions in a program. (1 mark)

SECTION 2: WORKED EXAMPLE — TRACE TABLE (guided)

Trace this pseudocode for the inputs 3 and 5:

$\text{num1} \leftarrow \text{INPUT}$

$\text{num2} \leftarrow \text{INPUT}$

$\text{total} \leftarrow 0$

WHILE $\text{num1} \leq \text{num2}$

$\text{total} \leftarrow \text{total} + \text{num1}$

$\text{num1} \leftarrow \text{num1} + 1$

ENDWHILE

OUTPUT total

num1	num2	total	Condition ($\text{num1} \leq \text{num2}$)?
3	5	0	True
4	5	3	True
_____	_____	_____	_____
_____	_____	_____	_____

OUTPUT: _____

Now you try:

Trace this pseudocode for the input 4:

$n \leftarrow \text{INPUT}$

$f \leftarrow 1$

FOR $i \leftarrow 1$ TO n

$f \leftarrow f \times i$

ENDFOR

OUTPUT f

Complete the trace table and state the output. (4 marks)

SECTION 3: INDEPENDENT PRACTICE

Q1. Write pseudocode that asks the user to input 10 numbers and outputs the largest number entered. (4 marks)

Q2. Draw a flowchart for an algorithm that checks if a password is at least 8 characters long. (3 marks)

[Use: oval for start/end, rectangle for process, diamond for decision, parallelogram for input/output]

Q3. Write a Python function called `is_even` that takes a number as a parameter and returns `True` if the number is even, `False` otherwise. (3 marks)

Q4. A program stores 5 test scores in an array called `scores`. Write pseudocode to calculate and output the average score. (4 marks)

Q5. Describe the steps you would take to test a program that calculates the area of a rectangle. Include at least 3 different types of test data with reasons. (4 marks)

VERSION C: DATABASES, SQL & BOOLEAN LOGIC

For students whose gap analysis shows weakness in: SQL queries, database design, Boolean algebra, or logic circuits.

SECTION 1: KNOWLEDGE CHECK (5 marks)

18. Write an SQL query to select the name and price from a table called `Products` where price is greater than 100. (1 mark)

2. What is the purpose of a primary key? (1 mark)

19. Draw the symbol for a NOT gate and state its truth table. (1 mark)

20. Simplify this Boolean expression: $A \text{ AND } (A \text{ OR } B)$ (1 mark)

21. State the purpose of normalisation in database design. (1 mark)

SECTION 2: WORKED EXAMPLE — SQL QUERY (guided)

Tables:

`Customers` (`CustomerID`, `Name`, `City`, `Email`)

`Orders` (`OrderID`, `CustomerID`, `Product`, `Quantity`, `OrderDate`)

Question: List the names of all customers from 'Shenzhen' who ordered more than 5 items, showing the product name and quantity.

Step 1: Which tables do we need? _____

Step 2: What is the linking column? _____

Step 3: Which columns do we SELECT? _____

Step 4: What conditions do we need in WHERE? _____

Complete the query:

```
SELECT Customers.Name, Orders.Product, Orders.Quantity
FROM Customers
_____ ON Customers.CustomerID = Orders.CustomerID
WHERE Customers.City = _____ AND Orders.Quantity > _____;
```

Now you try:

Write an SQL query to find all products ordered in March 2024, showing the product name and the customer's email. (3 marks)

SECTION 3: INDEPENDENT PRACTICE

Q1. A database has tables: Students(StudentID, Name, Class) and Grades(GradeID, StudentID, Subject, Mark). Write an SQL query to display each student's name and their mark in Mathematics. (3 marks)

Q2. Draw a logic circuit for the Boolean expression: $Q = (A \text{ AND } B) \text{ OR NOT } C$
Label all inputs, gates, and the output. (3 marks)

Q3. Complete this truth table for $Q = (A \text{ AND } B) \text{ OR NOT } C$: (4 marks)

A	B	C	NOT C	A AND B	Q
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

Q4. Simplify this Boolean expression using Boolean algebra laws. Show each step and state the law used:

$X = (A \text{ AND } B) \text{ OR } (A \text{ AND NOT } B)$ (3 marks)

Q5. Explain the difference between a flat-file database and a relational database. Give one advantage of using a relational database. (3 marks)

--- ANSWER KEY: WORKSHEET 11A ---

ANSWER KEY: WORKSHEET 11A — ALL VERSIONS

VERSION A ANSWERS: DATA REPRESENTATION & HARDWARE (25 marks total)

SECTION 1: KNOWLEDGE CHECK (5 marks)

22. 11001101 binary = $128+64+8+4+1 = 205$ denary (1 mark)

23. 9F hex = $(9 \times 16) + 15 = 159$ denary (1 mark)

24. Lossy: JPEG (1). Lossless: RLE / Run-Length Encoding (1).

Accept any valid examples. (1 mark total)

25. The ALU (Arithmetic Logic Unit) performs calculations (arithmetic) and logical comparisons (logic operations). (1 mark)

26. RAM is volatile (loses data when powered off), ROM is non-volatile (keeps data when powered off). OR RAM is read-write, ROM is read-only.

Must have one valid difference. (1 mark)

SECTION 2: WORKED EXAMPLE (5 marks)

Guided fill-in:

Step 2: 24

Step 3: $800 \times 600 \times 24 = 11,520,000$ bits

Step 4: $11,520,000 \div 8 = 1,440,000$ bytes

Step 5: $1,440,000 \div 1024 = 1406.25$ KB (or ≈ 1406 KB)

(1 mark per correct step = 4 marks for guided; method marks awarded)

Independent:

$1024 \times 768 \times 16 = 12,582,912$ bits

$\div 8 = 1,572,864$ bytes

$\div 1024 = 1536$ KB (or 1.5 MB)

(1 mark formula, 1 mark substitution, 1 mark correct answer)

SECTION 3: INDEPENDENT PRACTICE

Q1. Hex conversions (3 marks)

a) E3 (1)

b) ACF (1)

c) 1A85 (1)

Common error: grouping from left not right — watch for this!

Q2. Sound file calculation (4 marks)

Sampling rate = 44.1 kHz = 44,100 Hz

Duration = 3 minutes = 180 seconds

File size = $44,100 \times 16 \times 180 = 127,008,000$ bits

$\div 8 = 15,876,000$ bytes

$\div 1024 = 15,503.90625$ KB

$\div 1024 = 15.14$ MB (accept 15.1–15.2 MB)

(1 mark formula, 1 mark unit conversions, 1 mark calculation, 1 mark final answer)

Q3. RLE explanation (3 marks)

— RLE works by storing runs of the same colour as a single value + count (1)

— R R R R becomes R4 (or equivalent notation) (1)

— G B B B B W W becomes G1 B5 W2 (1)

— This reduces the total amount of data stored (1)

Max 3 marks

Q4. CPU diagram (4 marks)

- Correct boxes: ALU, CU, registers/cache, buses (2 marks)
- Arrows showing data flow between components (1 mark)
- Labels for address bus, data bus, control bus (1 mark)

Accept any clear, labelled diagram.

Q5. Cache memory (3 marks)

- Cache is fast memory between CPU and RAM (1)
- Stores frequently used data/instructions for quick access (1)
- Larger cache means more data can be stored there, reducing need to access slower RAM, improving performance (1)

VERSION B ANSWERS: PROGRAMMING & ALGORITHMS (24 marks total)

SECTION 1: KNOWLEDGE CHECK (5 marks)

27. num ← INPUT

IF num > 0 THEN

 OUTPUT "Positive"

ELSE

 OUTPUT "Not positive"

ENDIF (1 mark — syntax and logic correct)

28. x = 10 (0+1+2+3+4 = 10) (1 mark)

29. Any two from: normal (valid), boundary/extreme, erroneous/abnormal.

Example required for each. (1 mark total)

30. MOD gives the remainder after division. Example: 10 MOD 3 = 1 (1 mark)

31. Any one: reusable code, easier to debug, easier to maintain, reduces code duplication. (1 mark)

SECTION 2: WORKED EXAMPLE (7 marks)

Guided trace table:

Row 3: 5, 5, 7, True

Row 4: 6, 5, 12, False (loop ends)

OUTPUT: 12

(1 mark per correct row, 1 mark output)

Independent trace:

n=4, f=1

i=1: f=1×1=1

i=2: f=1×2=2

i=3: f=2×3=6

i=4: f=6×4=24

OUTPUT: 24

(1 mark per row, 1 mark output = 4 marks)

This calculates 4! (factorial)

SECTION 3: INDEPENDENT PRACTICE

Q1. Largest number pseudocode (4 marks)

largest ← 0

FOR count ← 1 TO 10

num ← INPUT

IF num > largest THEN

largest ← num

ENDIF

ENDFOR

OUTPUT largest

(1 mark loop, 1 mark input, 1 mark IF, 1 mark output/update)

Q2. Password length flowchart (3 marks)

— Start → Input password → Is length >= 8? → Yes: Output "Valid" / End

→ No: Output "Too short" / End

(1 mark each: correct symbols, correct logic, correct labels)

Q3. Python function (3 marks)

```
def is_even(number):
```

```
    if number % 2 == 0:
```

```
        return True
```

```
    else:
```

```
        return False
```

(1 mark def/name/parameter, 1 mark condition, 1 mark return)

Q4. Average score pseudocode (4 marks)

total ← 0

FOR i ← 0 TO 4

total ← total + scores[i]

ENDFOR

average ← total / 5

OUTPUT average

(1 mark total initialisation, 1 mark loop, 1 mark accumulation, 1 mark average/output)

Q5. Testing description (4 marks)

— Normal: width=10, height=5, expected area=50 (1 + reason 1)

- Boundary: width=0, height=5, expected area=0 (1 + reason 1)
- Erroneous: width="abc", height=5, should handle error (1 + reason 1)
- Max 4 marks (need 3 types with valid reasons)

VERSION C ANSWERS: DATABASES, SQL & BOOLEAN (24 marks total)

SECTION 1: KNOWLEDGE CHECK (5 marks)

32. SELECT Name, Price FROM Products WHERE Price > 100; (1 mark)
33. A primary key uniquely identifies each record in a table. (1 mark)
34. NOT gate: triangle with circle at output. Truth table: IN 0 → OUT 1, IN 1 → OUT 0. (1 mark for symbol + truth table)

4. A (absorption law: $A \text{ AND } (A \text{ OR } B) = A$) (1 mark)

35. Normalisation removes data redundancy/duplication and prevents data inconsistency by organising data into related tables. (1 mark)

SECTION 2: WORKED EXAMPLE (5 marks)

Step 1: Customers and Orders

Step 2: CustomerID

Step 3: Customers.Name, Orders.Product, Orders.Quantity

Step 4: Customers.City = 'Shenzhen' AND Orders.Quantity > 5

Complete query:

INNER JOIN Orders

'Shenzhen'

5

(1 mark per blank)

Independent SQL:

SELECT Orders.Product, Customers.Email

FROM Customers

INNER JOIN Orders ON Customers.CustomerID = Orders.CustomerID

WHERE Orders.OrderDate >= '2024-03-01' AND Orders.OrderDate <= '2024-03-31';

(1 mark SELECT, 1 mark JOIN correct, 1 mark WHERE correct)

SECTION 3: INDEPENDENT PRACTICE

Q1. SQL query (3 marks)

SELECT Students.Name, Grades.Mark

FROM Students

INNER JOIN Grades ON Students.StudentID = Grades.StudentID

WHERE Grades.Subject = 'Mathematics';

(1 mark SELECT, 1 mark JOIN, 1 mark WHERE)

Q2. Logic circuit (3 marks)

- A and B go into AND gate (1)
- C goes into NOT gate (1)
- Outputs of AND and NOT go into OR gate → Q (1)

Award marks for correct structure even if drawing is rough.

Q3. Truth table (4 marks)

A	B	C	NOT C	A AND B	Q
0	0	0	1	0	1
0	0	1	0	0	0
0	1	0	1	0	1
0	1	1	0	0	0
1	0	0	1	0	1
1	0	1	0	0	0
1	1	0	1	1	1
1	1	1	0	1	1

(0.5 marks per correct row, 8 rows = 4 marks)

Q4. Boolean simplification (3 marks)

$$X = (A \text{ AND } B) \text{ OR } (A \text{ AND NOT } B)$$

$$X = A \text{ AND } (B \text{ OR NOT } B) \quad \leftarrow \text{distributive law (1)}$$

$$X = A \text{ AND } 1 \quad \leftarrow B \text{ OR NOT } B = 1 \text{ (complement law) (1)}$$

$$X = A \quad \leftarrow A \text{ AND } 1 = A \text{ (identity law) (1)}$$

Q5. Flat-file vs relational (3 marks)

- Flat-file: all data in one table (1)
- Relational: data in multiple linked tables (1)
- Advantage: reduced data redundancy / no duplication / data consistency / easier to update / more efficient storage (any 1)

--- WORKSHEET 11B ---

WORKSHEET 11B: PERSONAL REVISION PLAN

Name: _____ Group (A/B/C): _____ Date: _____

MY SITUATION

- Days until Mock 4 (Paper 1): _____
- Days until Mock 5 (Paper 2): _____
- My current average: _____ / 75 (_____%)
- My target: _____ / 75 (_____%)
- I need to improve by: _____ marks

MY REVISION SCHEDULE (next 2 weeks)

Day	Date	Topic (EN)	Topic (中)	Time	Activity	Done?
-----	------	------------	-----------	------	----------	-------

			文)			
Mon						[]
Tue						[]
Wed						[]
Thu						[]
Fri						[]
Sat						[]
Sun						[]
Mon						[]
Tue						[]
Wed						[]
Thu						[]
Fri						[]

SUGGESTED ACTIVITIES (choose from this list or add your own):

- ☐ Complete revision worksheet questions
- ☐ Write flashcards for key terms (EN + 中文)
- ☐ Practise past paper questions (timed)
- ☐ Draw a mind map of a topic
- ☐ Explain a concept to a family member
- ☐ Watch a revision video and make notes
- ☐ Create a pseudocode/Pythons cheat sheet
- ☐ Practise SQL queries using an online tool

WHAT I LEARNED TODAY (from my revision group):

WHAT I STILL NEED HELP WITH:

MY PROMISE TO MYSELF:

I, _____, commit to _____ minutes of revision per day until my mocks. My priority topic is _____.

Signed: _____ Date: _____

--- ANSWER KEY: WORKSHEET 11B ---

ANSWER KEY: WORKSHEET 11B — PERSONAL REVISION PLAN

This is a planning worksheet — there are no "correct" answers. Award marks for completion and quality of planning.

MARKING CRITERIA (10 marks):

- My Situation section completed accurately (2 marks)
 - Correct calculation of days until mocks (1)
 - Average and target calculated correctly (1)
- Revision Schedule completed (4 marks)

- At least 8 days planned (2)
- Each entry has topic AND activity specified (1)
- Plan is realistic (not 5 hours on one day!) (1)
- What I Learned / What I Need Help With (2 marks)
 - Both sections completed with specific, meaningful content (2)
- My Promise to Myself signed and dated (1 mark)
 - Includes specific time commitment and priority topic (1)
- Overall presentation and effort (1 mark)

TEACHER NOTES:

- Collect, review, and return with written feedback before next lesson
- Flag any unrealistic plans (e.g., "revise everything in 2 days")
- Check that priority topics match gap analysis data
- Students should keep this visible at home — stick on bedroom wall!
- Follow up: ask students about their plan at the start of each lesson

WEEK 12: TARGETED REVISION — PAPER 1 WEAK AREAS

OVERVIEW:

Week 12 focuses exclusively on Paper 1 topics. Lesson 12.1 provides

whole-class interactive revision on the most common weak topics identified from mock data. Lesson 12.2 provides differentiated targeted practice where students work on worksheets specific to their personal weak areas. The three worksheet sets (A, B, C) allow every student to practise precisely what they need most.

TYPICAL PAPER 1 WEAK AREAS (adapt based on your class data):

36. File size calculations (sound and images)
37. Lossy vs lossless compression — explaining HOW, not just naming
38. CPU architecture and the fetch-execute cycle
39. Network hardware functions (router, switch, NIC)
40. Security threats — describing methods AND prevention
41. Emerging technologies — applying knowledge to new contexts

LESSON 12.1 — WHOLE-CLASS PAPER 1 REVISION: COMMON WEAK TOPICS

Topic: Interactive Revision of Class Paper 1 Weak Areas

Duration: 40 min | Lesson Type: Interactive Revision | Syllabus Ref: 1.1–1.6

Learning Objective:

By the end of this lesson, students will be able to answer questions on their class's most common Paper 1 weak topics with increased accuracy, using mini-whiteboards and quick-fire questioning to build confidence.

Key Vocabulary:

- quick-fire (快速提问 kuàisù tíwèn) — fast questions with short answers
- mini-whiteboard (小白板 xiǎo báibǎn) — small board for writing answers
- working (计算过程 jìsuàn guòchéng) — showing how you got the answer
- common misconception (常见误解 chángjiàn wùjiě) — mistake many people make
- recall (回忆 huíyì) — remembering facts

Lesson Structure (40 Minutes):

0–3 min | DO NOW

Each student receives a mini-whiteboard and pen. Teacher displays 3 questions on the board — one from each of the top 3 class weak topics. Students have 2 minutes to write answers on whiteboards and hold up. Teacher scans responses — notes patterns but does not correct yet.

3–12 min | DIRECT INSTRUCTION — TOPIC 1: FILE SIZE CALCULATIONS

Teacher addresses the #1 weak topic (typically file sizes):

— FORMULA REMINDER displayed large:

Image: $\text{width} \times \text{height} \times \text{colour depth (bits)} \rightarrow \div 8 \rightarrow \div 1024 \rightarrow \div 1024$

Sound: $\text{sampling rate} \times \text{duration} \times \text{sample depth (bits)} \rightarrow \div 8 \rightarrow \div 1024 \rightarrow \div 1024$

— COMMON MISCONCEPTION: "Always check units! kHz $\rightarrow$ Hz, minutes $\rightarrow$ seconds!"

— WORKED EXAMPLE on board: Sound file at 22.05 kHz, 8-bit, 2 minutes

— Show every step of working — students copy onto whiteboards

Students then do 2 quick-fire calculations independently:

Q1: Image 640×480, 16-bit colour. Size in KB?

Q2: Sound 44.1 kHz, 16-bit, 30 seconds. Size in MB?

Hold up answers — immediate feedback.

12–22 min | DIRECT INSTRUCTION — TOPIC 2: COMPRESSION

Teacher addresses compression weaknesses:

— "Lossy vs lossless is not just naming — you must explain HOW it works"

— RLE modelled with pixel row: B B B W W W W B B $\rightarrow$ B3 W4 B2

— Lossy (JPEG): permanently removes data humans are less sensitive to

— Lossless (RLE): stores repeated values as count + value, fully reversible

Quick-fire: "I have a 24-bit image with lots of solid blue sky. Which

compression and why?" $\rightarrow$ Lossless (RLE works well on solid colours) AND

lossy (JPEG works well on photos). Accept either with valid reason.

22–32 min | DIRECT INSTRUCTION — TOPIC 3: CPU / NETWORKS / SECURITY

Teacher addresses remaining weak topics (rotate based on class data):

- CPU: Fetch-Decode-Execute cycle as a flow diagram
MAR → address bus → memory → data bus → MDR → CIR → decode → execute
- Network hardware: Router (between networks), Switch (within LAN),
NIC (connects device to network)
- Security: Phishing (fake emails → verify sender), Malware (install
antivirus), SQL injection (input validation), DDoS (firewall/filtering)

Quick-fire questions interspersed every 2-3 minutes.

32-35 min | PLENARY

"3-2-1" on whiteboards:

- 3 facts you remembered today
- 2 things you need to revise more
- 1 question you still have

Teacher collects and reads aloud (anonymously) to address common themes.

35-40 min | HOMEWORK SET

Distribute Worksheet 12A (differentiated by weak area). Students receive either Set A (Data Representation), Set B (Hardware/Software), or Set C (Networks/Internet/Security) based on their gap analysis.

HOMEWORK: Complete your assigned worksheet set. Self-mark using the answer key at the back. Bring any questions to next lesson.

EAL Support:

- All formulas displayed visually with diagrams, not just text
- Mini-whiteboard responses can be diagrams/numbers — not just English
- Teacher speaks slowly and repeats key points during direct instruction
- Bilingual word bank on display for all technical terms
- EAL students can write 3-2-1 in Chinese if preferred

Differentiation:

Emerging:

- Pre-printed formula cards on desk for reference
- Teacher or LSA sits nearby during quick-fire for immediate help
- Only required to attempt 2 of 3 quick-fire questions
- Sentence starters for compression explanations

Developing:

- Full participation in all quick-fire questions
- Encouraged to show all working even if answer is wrong (method marks)
- Scaffolding removed gradually

Secure:

- Full participation plus extension questions during quick-fire
- Asked to explain reasoning to class
- Support Emerging students during pair moments

Resources Needed:

- Mini-whiteboards (one per student) + pens + erasers
- Projected formulas and worked examples
- Common misconceptions poster on wall
- 3-2-1 plenary template on board
- Worksheet 12A — differentiated sets ready to distribute

LESSON 12.2 — TARGETED PAPER 1 PRACTICE: DIFFERENTIATED WORKSHEETS

Topic: Differentiated Paper 1 Revision Practice

Duration: 40 min | Lesson Type: Independent Practice | Syllabus Ref: 1.1–1.6

Learning Objective:

By the end of this lesson, each student will have completed targeted practice questions on their specific Paper 1 weak areas and self-marked their work, identifying any remaining gaps.

Key Vocabulary:

- self-mark (自评 zì píng) — marking your own work
- target area (目标领域 mùbiāo lǐngyù) — the specific area you need to work on
- remain (剩余 shèngyú) — still left after something
- remaining gap (剩余差距 shèngyú chājù) — topic still not fully understood
- circulate (巡视 xúnshì) — teacher moving around the room

Lesson Structure (40 Minutes):

0–3 min | DO NOW

Students enter and collect their assigned worksheet set (A, B, or C) from the labelled trays at the front. They also collect the answer key (folded over — NOT to be opened until after completing work).

Students write their name on the worksheet and read the instructions.

3–30 min | INDEPENDENT PRACTICE

Students work independently on their differentiated worksheet:

Set A students (Data Rep weak): binary/hex conversions, file size calculations, compression explanations, number system questions

Set B students (Hardware weak): CPU architecture, memory types, storage comparison, OS functions, utility software, F-E cycle

Set C students (Networks weak): network topologies, hardware, protocols, IP addressing, security threats, emerging tech

Teacher circulates continuously:

- Spends most time with students who scored lowest on these topics
- Answers individual questions
- Checks working is being shown (reminds about method marks)
- Encourages use of formula cards and reference materials

Students who finish early:

- Self-mark using answer key
- Start second worksheet from another set (if time)
- Create 3 flashcards for their weakest sub-topic

30–35 min | SELF-MARKING

All students open answer keys and self-mark their work:

- Tick correct answers in green pen
- Write correct working for any wrong answers in red pen
- Total their marks
- Write a 1-sentence reflection: "I still need help with..."

Teacher circulates to verify self-marking is accurate and address any disputes about mark allocation.

35–40 min | PLENARY + HOMEWORK

"Traffic light" self-assessment for each worksheet topic:

Green = I can do this now

Yellow = I'm getting there but need more practice

Red = I still don't understand this

Students show coloured cards. Teacher notes reds for follow-up.

HOMEWORK: For any "red" topics, find and complete 2 additional questions from the textbook or revision guide. Bring evidence to next lesson.

EAL Support:

- All worksheets have key terms in English and Chinese
- Formula cards available in both languages
- Students can ask questions in Chinese — teacher or bilingual peer translates the answer
- Answer keys show mark breakdown (not just final answer) to help EAL students understand what examiners expect

Differentiation:

Emerging:

- Receive most teacher attention during circulation
- May use notes and textbook during worksheet
- Shorter worksheet (core questions only — extension removed)

- LSA support if available

Developing:

- Full worksheet with occasional teacher check-in
- Encouraged to attempt ALL questions before checking answers
- Prompted to show all working

Secure:

- Full worksheet + extension questions
- Self-mark independently
- Then support classmates or attempt questions from a different set
- Extension: create a "common mistakes" tip sheet for their topic area

Resources Needed:

- Worksheet 12A — 3 differentiated sets (A, B, C), multiple copies
- Answer keys for all sets (folded, one per student)
- Formula reference cards
- Green and red pens for self-marking
- Traffic light cards (green/yellow/red)
- Textbooks/revision guides for additional practice

--- WORKSHEET 12A ---

WORKSHEET 12A: TARGETED PAPER 1 REVISION (Differentiated — 3 Sets)

Instructions: Complete the set assigned by your teacher. Show ALL working.

SET A: DATA REPRESENTATION & NUMBER SYSTEMS

For students weak in: binary, hexadecimal, file sizes, compression

Q1. Convert 1101 0110 binary to denary. Show your working. (2 marks)

Q2. Convert 247 denary to hexadecimal. Show your working. (2 marks)

Q3. Convert 1010 1101 1111 0001 binary directly to hexadecimal. (2 marks)

Q4. An image has a resolution of 1920×1080 pixels and uses 32-bit colour.

Calculate the file size in megabytes (MB). Show all working. (4 marks)

Q5. A 3-minute sound recording is made at a sampling rate of 48 kHz with

24-bit sample depth. Calculate the file size in MB. Show all working.

(4 marks)

REMEMBER: convert kHz to Hz, minutes to seconds!

Q6. Explain the difference between lossy and lossless compression.

For each type, give one example of a file format that uses it. (4 marks)

Q7. A bitmap image contains the following row of pixels:

R R R R R G G B B B B W W W R R

Show how run-length encoding (RLE) would compress this row.

Calculate the compression ratio (original size : compressed size).

(4 marks)

Q8. A photographer takes photos for a magazine cover. Explain why they might choose to save the images using lossy compression rather than lossless.

(3 marks)

SET B: HARDWARE, SOFTWARE & THE CPU

For students weak in: CPU, memory, storage, operating systems, fetch-execute

Q1. State the purpose of each of the following CPU components: (3 marks)

- a) ALU
- b) Control Unit
- c) Registers

Q2. Describe the fetch-execute cycle. Your answer should include the roles of the MAR, MDR, PC and CIR. (6 marks)

Q3. Explain the difference between RAM and ROM. For each, state one use in a computer system. (4 marks)

Q4. Explain why increasing the amount of cache memory can improve the performance of a computer. (3 marks)

Q5. A student is choosing between a magnetic hard disk drive (HDD) and a solid-state drive (SSD) for their new computer.

- a) Give two advantages of an SSD over an HDD. (2 marks)
- b) Give one advantage of an HDD over an SSD. (1 mark)
- c) The student decides to use both. Explain how they might use each type of storage. (2 marks)

Q6. Describe the purpose of the operating system. Give three different functions. (3 marks)

Q7. Explain what is meant by defragmentation and why it is necessary for magnetic hard disk drives but not for SSDs. (4 marks)

Q8. Draw a diagram showing the von Neumann architecture. Label the main components and the buses that connect them. (4 marks)

Q9. Explain the purpose of virtual memory. Describe one disadvantage of using it. (3 marks)

SET C: NETWORKS, INTERNET & SECURITY

For students weak in: LAN/WAN, topologies, network hardware, protocols, security threats, emerging technologies

Q1. Describe three differences between a LAN and a WAN. (3 marks)

Q2. Draw and describe the star network topology. Give one advantage and one disadvantage of using a star topology. (4 marks)

Q3. For each of the following network devices, describe its purpose: (4 marks)

- a) Router
- b) Switch
- c) Network Interface Card (NIC)
- d) Wireless Access Point

Q4. Describe the client-server and peer-to-peer network models. Give one advantage of each model. (4 marks)

Q5. Explain the purpose of the TCP/IP protocol suite. Describe the role of IP addresses in routing data across the internet. (4 marks)

Q6. Explain how a packet of data is transmitted across the internet.

Include the following in your answer: packet, header, router, IP address.
(4 marks)

Q7. Describe three security threats to computer systems. For each threat, explain one method of prevention. (6 marks)

Q8. A company stores customer data on a database server. Explain two methods the company can use to keep this data secure. (4 marks)

Q9. Explain how artificial intelligence (AI) and machine learning are being used in one application area you have studied. Discuss one benefit and one ethical concern. (4 marks)

--- ANSWER KEY: WORKSHEET 12A ---

ANSWER KEY: WORKSHEET 12A — ALL SETS

SET A ANSWERS: DATA REPRESENTATION (25 marks)

Q1. 1101 0110 binary to denary (2 marks)

$$128 + 64 + 16 + 4 + 2 = 214$$

(1 mark correct method — showing powers of 2, 1 mark correct answer)

Q2. 247 denary to hexadecimal (2 marks)

$$247 \div 16 = 15 \text{ remainder } 7$$

$$15 = F$$

Answer: F7

(1 mark method, 1 mark answer)

Q3. 1010 1101 1111 0001 to hex (2 marks)

$$1010 = A, 1101 = D, 1111 = F, 0001 = 1$$

Answer: ADF1

(1 mark grouping correct, 1 mark hex digits)

Q4. Image file size (4 marks)

$$1920 \times 1080 \times 32 = 66,355,200 \text{ bits}$$

$\div 8 = 8,294,400$ bytes

$\div 1024 = 8,100$ KB

$\div 1024 = 7.91$ MB (accept 7.9 MB or ≈ 8 MB)

(1 mark formula, 1 mark bits, 1 mark bytes/KB, 1 mark MB)

Q5. Sound file size (4 marks)

48 kHz = 48,000 Hz

3 minutes = 180 seconds

$48,000 \times 24 \times 180 = 207,360,000$ bits

$\div 8 = 25,920,000$ bytes

$\div 1024 = 25,312.5$ KB

$\div 1024 = 24.72$ MB (accept 24.7 MB)

(1 mark unit conversions, 1 mark formula, 1 mark bits, 1 mark MB)

Q6. Lossy vs lossless (4 marks)

Lossy: permanently removes data (1), cannot recover original (1),
example JPEG/MP3 (1). Max 2 marks.

Lossless: no data lost (1), fully reversible (1), stores repeated data
efficiently e.g. RLE, ZIP (1). Max 2 marks.

Must have explanation + example for full marks.

Q7. RLE compression (4 marks)

Original: R R R R R G G B B B B W W W R R = 17 items

RLE: R5 G2 B6 W3 R2 = 10 items (accept any clear notation)

Compression ratio = 17:10 (accept 1.7:1)

(1 mark correct RLE, 1 mark original count, 1 mark compressed count,
1 mark ratio)

Q8. Lossy for magazine (3 marks)

- Lossy produces smaller file sizes (1)
- Photos have lots of detail where some data loss is not noticeable (1)
- Easier to upload/download/share (1)
- Suitable for web/print where perfect reproduction not critical (1)

Max 3 marks

SET B ANSWERS: HARDWARE, SOFTWARE & CPU (33 marks)

Q1. CPU components (3 marks)

- a) ALU — performs arithmetic calculations and logical comparisons (1)
- b) Control Unit — manages and coordinates all processor operations,
sends control signals (1)
- c) Registers — small, fast storage locations inside CPU holding

instructions/data being processed (1)

Q2. Fetch-execute cycle (6 marks)

- PC holds address of next instruction (1)
- Address copied to MAR (1)
- Address bus carries address to memory (1)
- Instruction at that address copied to MDR via data bus (1)
- Instruction copied from MDR to CIR (1)
- Instruction decoded by CU, then executed by ALU (1)
- PC incremented (1)

Max 6 marks. Accept diagram with labels.

Q3. RAM vs ROM (4 marks)

RAM: volatile (loses data when power off), read-write, stores running programs/data (1 + 1 for use e.g. running OS, open applications)

ROM: non-volatile (keeps data), read-only, stores boot instructions (1 + 1 for use e.g. BIOS/startup program)

Q4. Cache memory (3 marks)

- Cache is fast memory between CPU and RAM (1)
- Stores frequently accessed data/instructions (1)
- Larger cache = more data stored in fast access location = fewer accesses to slower RAM = faster processing (1)

Q5. HDD vs SSD (5 marks)

- a) SSD advantages (any 2): faster read/write speeds, no moving parts (more reliable, quieter, less heat), uses less power, lighter, faster boot time (2 marks)
- b) HDD advantage (1): cheaper per GB, larger capacities available, data recovery easier if failure
- c) Both used: SSD for OS and frequently used programs (fast access), HDD for large file storage (photos, videos) — cheaper for bulk (2)

Q6. Operating system functions (3 marks)

Any 3 from: memory management, processor management (scheduling), file management, security (user accounts), device management (drivers), user interface provision (3 marks)

Q7. Defragmentation (4 marks)

- Files stored in non-contiguous blocks become fragmented over time (1)
- Defragmentation rearranges files into contiguous blocks (1)
- Reduces read/write head movement on HDD → faster access (1)

- SSDs have no moving parts, no head movement, access time same regardless of location → defragmentation unnecessary + causes unnecessary wear on flash memory (1)

Q8. Von Neumann diagram (4 marks)

- CPU (ALU, CU, registers) connected to memory (RAM) via buses (2)
- Input and output devices shown (1)
- All three buses labelled: address bus, data bus, control bus (1)

Accept any clear, labelled diagram.

Q9. Virtual memory (3 marks)

- Uses part of HDD/SSD as temporary RAM when RAM is full (1)
- Allows more programs to run than RAM capacity (1)
- Disadvantage: much slower than RAM (1) OR increases HDD wear (1)

SET C ANSWERS: NETWORKS, INTERNET & SECURITY (37 marks)

Q1. LAN vs WAN (3 marks)

Any 3 valid differences:

- LAN covers small area (building), WAN covers large area (country/globe)
- LAN owned by one organisation, WAN uses shared infrastructure
- LAN typically faster/wired, WAN typically slower/uses leased lines
- LAN: private, WAN: public/private mix

(1 mark per valid difference)

Q2. Star topology (4 marks)

- Diagram: central device (switch/hub) with all nodes connected directly to it (1 mark clear diagram)
- Description: all devices connected to central point (1)
- Advantage: easy to add/remove devices, fault in one cable doesn't affect others, easy to diagnose faults (1)
- Disadvantage: if central device fails whole network fails, expensive cabling (more cable than bus) (1)

Q3. Network devices (4 marks)

- a) Router — forwards data between networks using IP addresses, finds best path (1)
- b) Switch — forwards data within LAN using MAC addresses, connects multiple devices (1)
- c) NIC — allows device to connect to network, has unique MAC address (1)
- d) WAP — allows wireless devices to connect to wired network (1)

Q4. Network models (4 marks)

Client-server: central server provides services/resources, clients request services, server manages security and data (1 + advantage 1)
Peer-to-peer: all devices equal, share resources directly, no central server (1 + advantage 1 e.g. easy to set up, no single point of failure)

Q5. TCP/IP (4 marks)

- TCP ensures reliable delivery, breaks data into packets, reassembles at destination, handles errors (1 + 1)
- IP handles addressing and routing, each device has unique IP address, routers use IP addresses to forward packets to correct destination (1 + 1)

Q6. Packet transmission (4 marks)

- Data split into packets (1)
- Each packet has header with destination IP address, sequence number, source IP (1)
- Packets travel through routers which read destination IP and forward to next router (1)
- Packets may take different routes, reassembled at destination using sequence numbers (1)

Q7. Security threats (6 marks)

Any 3 threats with prevention:

- Malware: install and update antivirus, don't download unknown files (2)
- Phishing: verify sender, don't click suspicious links, email filters (2)
- SQL injection: input validation, parameterised queries (2)
- DDoS: firewalls, rate limiting, over-provision bandwidth (2)
- Hacking: strong passwords, firewalls, encryption (2)

(1 mark threat description, 1 mark prevention per pair)

Q8. Data security methods (4 marks)

Any 2 methods:

- Encryption: data scrambled so unreadable without key (2)
- Access control: usernames/passwords, permission levels (2)
- Firewall: blocks unauthorised access (2)
- Regular backups: recover data if lost/stolen (2)
- Physical security: locked server room (2)

(2 marks per method — description + how it protects)

Q9. AI application (4 marks)

Any valid application: healthcare (diagnosis), transport (self-driving), finance (fraud detection), education (personalised learning), etc.

- Description of how AI/ML is used in chosen area (1)
- Benefit: e.g. faster diagnosis, reduced human error, 24/7 operation, handles large data (1 + 1 development)
- Ethical concern: job displacement, bias in training data, privacy, accountability, dependency (1)

Max 4 marks

--- WORKSHEET 12B ---

WORKSHEET 12B: PAPER 1 QUICK-FIRE PRACTICE

Name: _____ Date: _____ Set (A/B/C): _____

30 quick questions — test yourself against the clock! Target: 15 minutes.

- Q1. What is the denary value of 0001 0000?
- Q2. What is 255 in hexadecimal?
- Q3. How many bits are in 2 bytes?
- Q4. What is the formula for calculating image file size (in bits)?
- Q5. What does "lossy" compression lose?
- Q6. Name the three buses in the von Neumann architecture.
- Q7. What does the ALU do?
- Q8. What type of memory is RAM?
- Q9. What does "non-volatile" mean?
- Q10. What is the purpose of an operating system?
- Q11. Name two utility programs.
- Q12. What is defragmentation?
- Q13. What does LAN stand for?
- Q14. What device connects a LAN to the internet?
- Q15. What is the difference between a router and a switch?
- Q16. What does TCP stand for?
- Q17. What is an IP address used for?
- Q18. What is phishing?
- Q19. Name two ways to prevent malware infection.
- Q20. What is encryption?
- Q21. What is the difference between client-server and peer-to-peer?
- Q22. What does "bandwidth" mean?
- Q23. What is a MAC address?
- Q24. What is the purpose of a firewall?
- Q25. What is cloud computing?
- Q26. Name one emerging technology and one use.

Q27. What does "volatile" mean for memory?

Q28. What is cache memory?

Q29. Name three input devices.

Q30. What is the fetch-execute cycle?

SCORE: _____ / 30 Time taken: _____ minutes

--- ANSWER KEY: WORKSHEET 12B ---

ANSWER KEY: WORKSHEET 12B — QUICK-FIRE PRACTICE (30 marks)

Q1. 16

Q2. FF

Q3. 16 bits (2×8)

Q4. width $\times$ height $\times$ colour depth

Q5. Permanently removes some data (cannot be recovered)

Q6. Address bus, data bus, control bus

Q7. Arithmetic and logic operations/calculations and comparisons

Q8. Volatile (loses data when power off) + read-write

Q9. Keeps data even when power is turned off

Q10. Manages hardware/software resources (any valid function)

Q11. Any two: antivirus, defragmenter, disk cleanup, backup, compression,
firewall

Q12. Reorganising files on disk into contiguous blocks to speed up access

Q13. Local Area Network

Q14. Router (or modem/gateway)

Q15. Router connects networks; switch connects devices within same network

Q16. Transmission Control Protocol

Q17. Uniquely identifies a device on a network; used for routing packets

Q18. Fraudulent attempt to obtain sensitive information by disguising as
trustworthy entity (usually via email)

Q19. Any two: install antivirus, keep software updated, don't click unknown
links, don't download untrusted files, use firewall

Q20. Converting data into a code to prevent unauthorised access

Q21. Client-server: central server provides services to clients.

Peer-to-peer: all devices equal, share directly.

Q22. The maximum rate of data transfer across a network connection

Q23. Unique hardware identifier assigned to a NIC (Media Access Control)

Q24. Monitors and controls network traffic based on security rules;
blocks unauthorised access

- Q25. Delivering computing services (storage, processing) over the internet
- Q26. Any valid: AI (healthcare diagnosis), blockchain (secure transactions), IoT (smart homes), VR (training), quantum computing (research)
- Q27. Data is lost when power is removed
- Q28. Small, very fast memory between CPU and RAM storing frequently used data
- Q29. Any three: keyboard, mouse, scanner, microphone, webcam, touchscreen, barcode reader, sensors
- Q30. The cycle of fetching instructions from memory, decoding them, and executing them, repeated continuously

SCORING GUIDE:

27–30 = Excellent — secure knowledge

22–26 = Good — minor gaps to address

17–21 = Developing — several topics need revision

Below 17 = Emerging — significant revision needed; prioritise weak topics

WEEK 13: TARGETED REVISION — PAPER 2 WEAK AREAS

OVERVIEW:

Week 13 mirrors Week 12 but focuses exclusively on Paper 2 topics.

Lesson 13.1 provides whole-class interactive revision on the most common Paper 2 weak topics. Lesson 13.2 provides differentiated targeted practice with three worksheet sets covering: (A) Algorithms/Pseudocode, (B) Python Programming, and (C) Databases/SQL/Boolean. The 15-mark scenario question gets dedicated attention for students who need it.

TYPICAL PAPER 2 WEAK AREAS (adapt based on your class data):

- 42. Pseudocode writing — syntax errors, forgetting ENDFOR/ENDIF
- 43. Trace tables with nested loops or WHILE loops
- 44. SQL queries — especially JOIN and aggregate functions
- 45. Boolean logic circuits — drawing from expressions
- 46. The 15-mark scenario question — structured design and justification

LESSON 13.1 — WHOLE-CLASS PAPER 2 REVISION: COMMON WEAK TOPICS

Topic: Interactive Revision of Class Paper 2 Weak Areas

Duration: 40 min | Lesson Type: Interactive Revision | Syllabus Ref: 2.1–2.3

Learning Objective:

By the end of this lesson, students will be able to write pseudocode

with correct syntax, complete trace tables accurately, construct SQL queries with JOINS, and draw logic circuits from Boolean expressions.

Key Vocabulary:

- syntax (语法 yǔfǎ) — the rules of how to write code correctly
- trace table (追踪表 zhuīzōng biǎo) — a table showing variable values
- nested loop (嵌套循环 qiàntào xúnhuán) — a loop inside another loop
- aggregate function (聚合函数 jùhé hánshù) — COUNT, SUM, AVG, MAX, MIN
- scenario question (情景题 qíngjǐng tí) — a long question with context

Lesson Structure (40 Minutes):

0–3 min | DO NOW

Mini-whiteboard warm-up:

"Write the first line of pseudocode to declare a variable 'total' with value 0."

"Write the SQL keyword to combine two tables."

Students hold up — teacher scans for common errors.

3–15 min | DIRECT INSTRUCTION — PSEUDOCODE & TRACE TABLES

Teacher addresses the #1 weak topic (typically pseudocode writing):

— SYNTAX RULES displayed on board:

← (NOT =) for assignment

OUTPUT (NOT print)

INPUT (NOT input())

ENDIF, ENDFOR, ENDWHILE (must close every block)

FOR i ← 1 TO 10 (inclusive — goes 1,2,3,...,10)

WHILE condition ... ENDWHILE (condition checked before each iteration)

— COMMON ERROR: forgetting ENDFOR / ENDIF

Teacher shows a pseudocode snippet missing ENDFOR — students find the error on whiteboards

— WORKED EXAMPLE: Write pseudocode to input 5 numbers and find their sum

Teacher models on board, students copy:

total ← 0

FOR count ← 1 TO 5

num ← INPUT

total ← total + num

ENDFOR

OUTPUT total

Students then write pseudocode independently:

"Input numbers until the user enters -1. Count how many numbers

were entered."

Teacher shows a model answer, students self-check.

TRACE TABLE TECHNIQUE:

- One row per iteration
- One column per variable AND the condition
- Watch for: loop counter changes, accumulation variables, condition becoming false
- Worked example: teacher traces a WHILE loop on board with students copying into their books

15–25 min | DIRECT INSTRUCTION — SQL & BOOLEAN

Teacher addresses SQL JOINS (typically a major weak area):

- "JOIN requires a linking column — usually the primary key of one table matches the foreign key of the other"
- TEMPLATE displayed:

```
SELECT table1.column, table2.column  
FROM table1  
INNER JOIN table2 ON table1.key = table2.key  
WHERE condition;
```

- Worked example with real table data projected

Students do quick-fire: "Write the SQL to list all students and their grades in Mathematics" (tables projected).

BOOLEAN LOGIC CIRCUITS:

- Quick review: AND (both true), OR (at least one true), NOT (inverts)
- Drawing technique: inputs on left, gates in middle, output on right
- Worked example: Draw circuit for $Q = (A \text{ AND } B) \text{ OR NOT } C$
- Students draw on whiteboards, hold up

25–32 min | THE 15-MARK SCENARIO

Teacher introduces the 15-mark question structure:

- Usually: read a scenario, design a solution (pseudocode/program/DB), explain choices, justify decisions
- MARK BREAKDOWN (typical):
 - Design: appropriate structure chosen (3)
 - Technical accuracy: correct syntax/logic (5)
 - Explanation: WHY choices were made (4)
 - Quality: testing, validation, edge cases (3)
- STRATEGY: "Plan before you write. Spend 3 minutes thinking."

- Students see a model 15-mark answer with annotations showing where each mark comes from

32–35 min | PLENARY

"Pseudocode poker": Each student writes one line of pseudocode on a card. Cards are shuffled and read aloud. Class votes: valid syntax or error? If error, how to fix it?

35–40 min | HOMEWORK SET

Distribute Worksheet 13A (differentiated sets A, B, C).

HOMEWORK: Complete your assigned set. Self-mark. 15-mark question extension for those who need practice (optional but recommended for students who scored below 8/15 in mocks).

EAL Support:

- Pseudocode syntax rules displayed as a visual poster (permanent)
- SQL template cards on desks for reference
- Trace table format pre-drawn on whiteboard/sheet
- Teacher models slowly with gestures pointing to each part of code
- Bilingual pseudocode keyword list available

Differentiation:

Emerging:

- During pseudocode writing, provided with sentence/frame structures:
" _____ ← _____ " for assignment, "IF _____ THEN" for selection
- Trace table pre-started (first row filled in)
- SQL: given template with most keywords filled in (student adds table/column names)

Developing:

- Full participation with occasional scaffolding
- During 15-mark section, given a planning framework to structure answer

Secure:

- Full participation plus extension challenges throughout
- 15-mark practice with examiner-style mark scheme analysis
- Support peers during pair/group moments

Resources Needed:

- Mini-whiteboards and pens
- Projected syntax rules and SQL template
- Pseudocode poker cards
- Worksheet 13A — 3 differentiated sets
- 15-mark scenario example (annotated) for projection
- Planning framework handout for 15-mark questions

LESSON 13.2 — TARGETED PAPER 2 PRACTICE: DIFFERENTIATED WORKSHEETS

Topic: Differentiated Paper 2 Revision Practice

Duration: 40 min | Lesson Type: Independent Practice | Syllabus Ref: 2.1–2.3

Learning Objective:

By the end of this lesson, each student will have completed targeted practice on their specific Paper 2 weak areas, self-marked accurately, and identified any remaining topics requiring further revision.

Key Vocabulary:

- independent practice (独立练习 dúlì liànxí) — practising on your own
- self-mark (自评 zì píng) — marking your own answers
- remaining topic (剩余主题 shèngyú zhǔtí) — topic still needing work
- scenario (情景 qíngjǐng) — a situation or story setting a problem
- justify (证明 zhèngmíng) — give reasons for your choices

Lesson Structure (40 Minutes):

0–3 min | DO NOW

Students collect their assigned worksheet set (A, B, or C) and the corresponding 15-mark scenario practice sheet (if assigned). Teacher reminds: "Show all working. Use correct pseudocode syntax. Check your trace table row by row."

3–30 min | INDEPENDENT PRACTICE

Students work independently on their targeted worksheet:

Set A (Algorithms/Pseudocode): writing pseudocode, trace tables, flowchart interpretation, algorithm design

Set B (Python Programming): code reading, writing functions, debugging, code completion

Set C (Databases/SQL/Boolean): SQL queries, database design, truth tables, Boolean simplification, logic circuits

15-mark scenario: students assigned this work on a separate sheet with a structured planning template.

Teacher circulates:

- Prioritises students with lowest mock scores
- Checks pseudocode syntax (CAPS, $\leftarrow$ not $=$, ENDIF present)
- Verifies trace tables are being filled one row at a time
- Helps SQL students identify linking columns for JOINS
- Reads 15-mark plans before students start writing

30–35 min | SELF-MARKING

Students use answer keys to self-mark:

- Mark each question with ticks and marks
- Write corrections in a different colour
- Total the score
- Complete reflection: "My strongest topic today: _____ / My weakest: _____"

Teacher verifies marking accuracy.

35–40 min | PLENARY + HOMEWORK

Traffic light assessment (green/yellow/red) for each major topic.

Students with "red" on any topic sign up for a 5-minute teacher conference before next lesson.

HOMEWORK: Complete any unfinished questions. For "red" topics, complete 2 additional questions from revision guide.

EAL Support:

- All worksheets bilingual — key terms in English and Chinese
- Pseudocode keywords reference card available
- SQL syntax template card on desks
- Trace table templates pre-printed
- Answer keys show mark allocations clearly (students learn what examiners expect even if language is still developing)

Differentiation:

Emerging:

- Shorter worksheet (core only)
- Reference materials allowed (syntax cards, keyword lists)
- Teacher or LSA provides frequent check-ins
- 15-mark scenario: structured writing frame provided

Developing:

- Full worksheet
- Self-mark with teacher spot-checks
- 15-mark: planning template provided

Secure:

- Full worksheet + extension
- 15-mark: full independent practice with examiner-style self-assessment
- Support peers after completing own work
- Extension: write an examiner-style mark scheme for a peer's 15-mark answer

Resources Needed:

- Worksheet 13A — 3 differentiated sets (A, B, C)
- Worksheet 13B — 15-mark scenario practice (optional)
- Answer keys for all materials

- Self-marking pens (green/red)
- Traffic light cards
- Planning template for 15-mark questions
- Reference cards (pseudocode keywords, SQL template, Boolean laws)

--- WORKSHEET 13A ---

WORKSHEET 13A: TARGETED PAPER 2 REVISION (Differentiated — 3 Sets)

Instructions: Complete the set assigned by your teacher. Use correct syntax.

SET A: ALGORITHMS, PSEUDOCODE & TRACE TABLES

Q1. State three rules for writing correct pseudocode. (3 marks)

Q2. Trace the following pseudocode using the trace table provided: (5 marks)

$x \leftarrow 5$

$y \leftarrow 1$

WHILE $x > 0$

$y \leftarrow y \times x$

$x \leftarrow x - 1$

ENDWHILE

OUTPUT y

| x | y | $x > 0?$ |

OUTPUT: _____

What does this algorithm calculate? _____

Q3. Write pseudocode for an algorithm that: (4 marks)

- Asks the user to enter their age
- If age is 18 or over, outputs "Adult"
- If age is 13–17, outputs "Teenager"
- Otherwise, outputs "Child"

Q4. Write pseudocode that inputs 20 test scores and counts how many are 50 or above (pass) and how many are below 50 (fail). Output both counts. (5 marks)

Q5. Study this flowchart description: (4 marks)

Start → Input number → Is number MOD 2 = 0?

→ YES: Output "Even" → Stop

→ NO: Output "Odd" → Stop

- Draw the flowchart using correct symbols.
- Write equivalent pseudocode.

Q6. Write pseudocode to find the smallest number in an array of 100 numbers called data. (4 marks)

SET B: PYTHON PROGRAMMING

Q1. What is the output of this Python code? (2 marks)

```
for i in range(3):  
    print(i * 2)
```

Q2. Find and correct THREE errors in this Python function: (3 marks)

```
def calculate_area(length, width)  
    area = Length * width  
    print "The area is " + area  
    Return area
```

Q3. Write a Python function called grade_mark that takes a mark (0-100) and returns: (5 marks)

- "A" if mark >= 80
- "B" if mark >= 65
- "C" if mark >= 50
- "F" if mark < 50

Include input validation to check the mark is between 0 and 100.

Q4. Write Python code to create an empty list called temperatures, then use a loop to input 7 temperature readings, store them in the list, and calculate the average. Output the average rounded to 1 decimal place. (5 marks)

Q5. What is the purpose of a test plan? Describe three types of test data with examples. (4 marks)

Q6. The following function should check if a password is strong (at least 8 characters and contains at least one number). Complete the missing code. (6 marks)

```
def is_strong_password(password):  
    has_number = False  
    if len(password) < ____:  
        return ____  
    for char in password:  
        if ____:  
            has_number = ____  
    if has_number:  
        return ____  
    else:  
        return ____
```

SET C: DATABASES, SQL & BOOLEAN LOGIC

Q1. Write an SQL query to display all records from a table called Employees where the department is 'IT', ordered by salary from highest to lowest.

(2 marks)

Q2. Tables: Customers(CustomerID, Name, City) and Orders(OrderID, CustomerID, Amount, OrderDate)

Write SQL queries to: (6 marks)

- a) List all customers and their order amounts (JOIN).
- b) Find the total amount spent by each customer (GROUP BY).
- c) Find customers who have NOT placed any orders (use LEFT JOIN or NOT IN).

Q3. A database stores student information in a single table with these fields: StudentID, Name, Class, Subject1, Subject2, Subject3, Teacher1, Teacher2, Teacher3. Explain why this design is not normalised and describe how you would improve it to Third Normal Form (3NF). (5 marks)

Q4. a) Draw a logic circuit for the Boolean expression: $Q = A \text{ AND } (B \text{ OR NOT } C)$

(3 marks)

b) Complete the truth table for this expression: (4 marks)

A	B	C	NOT C	B OR NOT C	Q
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			

Q5. Simplify using Boolean algebra. Show each step:

$X = A \text{ AND } (\text{NOT } A \text{ OR } B)$ (3 marks)

Q6. A school database needs to track which students are enrolled in which courses. Draw the entity-relationship diagram showing the tables, primary keys, foreign keys, and relationships needed. Explain your design.

(4 marks)

--- ANSWER KEY: WORKSHEET 13A ---

ANSWER KEY: WORKSHEET 13A — ALL SETS

SET A ANSWERS: ALGORITHMS & PSEUDOCODE (25 marks)

Q1. Pseudocode rules (3 marks)

Any 3 from:

- Use $\leftarrow$ for assignment (not =)
- Keywords in CAPS (INPUT, OUTPUT, IF, THEN, ELSE, ENDIF, FOR, TO, ENDFOR, WHILE, ENDWHILE)
- Every IF must have ENDIF, every FOR must have ENDFOR, etc.
- Use meaningful variable names
- One instruction per line
- OUTPUT not print, INPUT not input

Q2. Trace table (5 marks)

x	y	x > 0?
5	1	True
4	5	True
3	20	True
2	60	True
1	120	True
0	120	False

OUTPUT: 120

This calculates factorial ($5! = 120$)

(1 mark per correct row, 1 mark output, 1 mark identification)

Q3. Age pseudocode (4 marks)

```

age ← INPUT
IF age >= 18 THEN
    OUTPUT "Adult"
ELSE IF age >= 13 THEN
    OUTPUT "Teenager"
ELSE
    OUTPUT "Child"
ENDIF

```

(1 mark input, 1 mark IF structure, 1 mark correct conditions,
1 mark correct outputs — deduct 0.5 for missing ENDIF)

Q4. Pass/fail count (5 marks)

```

pass ← 0
fail ← 0
FOR i ← 1 TO 20
    score ← INPUT
    IF score >= 50 THEN
        pass ← pass + 1
    ELSE
        fail ← fail + 1
    ENDIF
ENDFOR

```

ELSE

fail ← fail + 1

ENDIF

ENDFOR

OUTPUT pass, fail

(1 mark variables, 1 mark loop, 1 mark input, 1 mark IF, 1 mark output)

Q5. Flowchart and pseudocode (4 marks)

a) Flowchart: correct symbols (1) — oval start/end, parallelogram input, diamond decision, rectangle process/output

b) Pseudocode:

number ← INPUT

IF number MOD 2 = 0 THEN

OUTPUT "Even"

ELSE

OUTPUT "Odd"

ENDIF (1 mark)

(2 marks for flowchart, 2 marks for pseudocode)

Q6. Smallest number (4 marks)

smallest ← data[0]

FOR i ← 1 TO 99

IF data[i] < smallest THEN

smallest ← data[i]

ENDIF

ENDFOR

OUTPUT smallest

(1 mark initialisation, 1 mark loop, 1 mark comparison, 1 mark output)

SET B ANSWERS: PYTHON PROGRAMMING (25 marks)

Q1. Output (2 marks)

0

2

4

(1 mark each correct line — deduct 0.5 per error)

Q2. Three errors (3 marks)

Error 1: Missing colon after function definition → def calculate_area(length, width):

Error 2: Length should be lowercase → length (case sensitive)

Error 3: Cannot concatenate string and int → print("The area is", area)

OR `print("The area is " + str(area))`

Error 4: Return should be lowercase → `return`

(1 mark per error identified and corrected, max 3)

Q3. Grade function (5 marks)

```
def grade_mark(mark):  
    if mark < 0 or mark > 100:  
        return "Invalid mark"  
    if mark >= 80:  
        return "A"  
    elif mark >= 65:  
        return "B"  
    elif mark >= 50:  
        return "C"  
    else:  
        return "F"
```

(1 mark function definition, 1 mark validation, 1 mark correct grade boundaries, 1 mark elif structure, 1 mark returns correct grades)

Q4. Temperature list (5 marks)

```
temperatures = []  
for i in range(7):  
    temp = float(input("Enter temperature: "))  
    temperatures.append(temp)  
average = sum(temperatures) / len(temperatures)  
print(round(average, 1))
```

(1 mark empty list, 1 mark loop, 1 mark input+append, 1 mark average calc, 1 mark rounded output)

Q5. Test plan (4 marks)

- Purpose: to check program works correctly, find errors, verify outputs for different inputs (1)
- Normal: valid input expected to work, e.g. mark=75 for grade B (1)
- Boundary: at limit, e.g. mark=50 (pass boundary), mark=0, mark=100 (1)
- Erroneous: invalid input, e.g. mark=-5 or mark=150 (1)

Must have type + example + reason for each.

Q6. Complete function (6 marks)

```
def is_strong_password(password):  
    has_number = False
```

```

if len(password) < 8:
    return False
for char in password:
    if char.isdigit():
        has_number = True
if has_number:
    return True
else:
    return False

```

(1 mark each blank: 8, False, char.isdigit(), True, True, False)

SET C ANSWERS: DATABASES, SQL & BOOLEAN (27 marks)

Q1. SQL query (2 marks)

```

SELECT * FROM Employees
WHERE Department = 'IT'
ORDER BY Salary DESC;

```

(1 mark WHERE, 1 mark ORDER BY DESC)

Q2. SQL queries (6 marks)

a) SELECT Customers.Name, Orders.Amount

FROM Customers

INNER JOIN Orders ON Customers.CustomerID = Orders.CustomerID;

(1 mark SELECT, 1 mark JOIN correct)

b) SELECT Customers.Name, SUM(Orders.Amount) AS Total

FROM Customers

INNER JOIN Orders ON Customers.CustomerID = Orders.CustomerID

GROUP BY Customers.CustomerID, Customers.Name;

(1 mark SUM, 1 mark GROUP BY)

c) SELECT Name FROM Customers

WHERE CustomerID NOT IN (SELECT CustomerID FROM Orders);

OR: LEFT JOIN with NULL check

(1 mark method, 1 mark correct syntax)

Q3. Normalisation (5 marks)

— Not normalised: repeating groups (Subject1, Subject2, Subject3),
data redundancy, update anomalies (1)

— 1NF: Separate into Students(StudentID, Name, Class) and
Enrolments(StudentID, Subject, Teacher) — no repeating groups (1)

— 2NF: Already in 2NF if 1NF (no partial dependencies) (1)

- 3NF: Separate Subjects table: Subjects(SubjectID, SubjectName, Teacher)
- Enrolments(StudentID, SubjectID) — removes transitive dependency (1)
- Explanation of benefit: no data redundancy, easier to update, no inconsistency (1)

Q4. Logic circuit and truth table (7 marks)

- a) Circuit: B and C go into NOT C, then B and NOT C into OR gate, A and output of OR go into AND gate → Q (3 marks)

b) Truth table:

A	B	C	NOT C	B OR NOT C	Q
0	0	0	1	1	0
0	0	1	0	0	0
0	1	0	1	1	0
0	1	1	0	1	0
1	0	0	1	1	1
1	0	1	0	0	0
1	1	0	1	1	1
1	1	1	0	1	1

(4 marks — 0.5 per row)

Q5. Boolean simplification (3 marks)

$$X = A \text{ AND } (\text{NOT } A \text{ OR } B)$$

$$X = (A \text{ AND NOT } A) \text{ OR } (A \text{ AND } B) \leftarrow \text{distributive law (1)}$$

$$X = 0 \text{ OR } (A \text{ AND } B) \leftarrow A \text{ AND NOT } A = 0 \text{ (complement) (1)}$$

$$X = A \text{ AND } B \leftarrow \text{identity law (1)}$$

Q6. ER diagram (4 marks)

- Students table: StudentID (PK), Name, ... (1)
- Courses table: CourseID (PK), CourseName, ... (1)
- Enrolments table: StudentID (FK), CourseID (FK) — linking table for many-to-many (1)
- Explanation: Many students can enrol in many courses, so linking table needed with foreign keys (1)

--- WORKSHEET 13B ---

WORKSHEET 13B: THE 15-MARK SCENARIO QUESTION

Name: _____ Date: _____

SCENARIO: SCHOOL LIBRARY SYSTEM

Victoria Park Academy is developing a digital library system. Students can borrow books, reserve books, and search the catalogue. The system needs to:

- Store book information: title, author, ISBN, genre, total copies, copies available
 - Store student borrower information: student ID, name, class, email
 - Track loans: which student borrowed which book, borrow date, due date, return date, overdue status
 - Send automatic email reminders when books are overdue
 - Generate reports: most popular books, overdue books, borrowing by class
- The system should be menu-driven. When a student logs in, they see options:

1. Search for a book

2. Borrow a book

3. Return a book

4. View my loans

5. Exit

QUESTION (15 marks)

- (a) Design a database for this system. Show the tables you would create, with field names, data types, and primary keys. Explain how the tables are related. (5 marks)
- (b) Write pseudocode for option 2 (Borrow a book). Your pseudocode should:
- Input the book title or ISBN
 - Check if the book is available
 - If available, reduce copies available by 1 and record the loan
 - If not available, allow the student to reserve it
 - Output appropriate messages
- (6 marks)
- (c) Explain two features you would include to make the system secure. Give a reason for each. (2 marks)
- (d) Describe how you would test the borrow book function. Include at least two different types of test data with reasons. (2 marks)

PLANNING TEMPLATE (use this before you write your answer)

Part (a) — Database design:

Tables I need: _____

Primary keys: _____

Foreign keys: _____

Relationships: _____

Part (b) — Pseudocode plan:

Inputs needed: _____

Processes: _____

Outputs: _____

Special cases (no copies, book not found): _____

Part (c) — Security features:

Feature 1: _____ Reason: _____

Feature 2: _____ Reason: _____

Part (d) — Testing:

Test 1 type: _____ Data: _____ Reason: _____

Test 2 type: _____ Data: _____ Reason: _____

--- ANSWER KEY: WORKSHEET 13B ---

ANSWER KEY: WORKSHEET 13B — 15-MARK SCENARIO (15 marks)

MARK SCHEME

Part (a) — Database design (5 marks)

Tables (3 marks):

Books: ISBN (PK, STRING), Title (STRING), Author (STRING), Genre (STRING),
TotalCopies (INTEGER), AvailableCopies (INTEGER) (1 mark)

Students: StudentID (PK, STRING), Name (STRING), Class (STRING),
Email (STRING) (1 mark)

Loans: LoanID (PK, INTEGER/STRING), ISBN (FK), StudentID (FK),
BorrowDate (DATE), DueDate (DATE), ReturnDate (DATE),
Overdue (BOOLEAN) (1 mark)

Relationships (1 mark):

- Books to Loans: one-to-many (one book can have many loans) (0.5)
- Students to Loans: one-to-many (one student can have many loans) (0.5)
- Linked by foreign keys (ISBN and StudentID in Loans table)

Explanation (1 mark):

- Separating into three tables avoids data redundancy / in 3NF /
prevents update anomalies / each piece of data stored once (1)

Part (b) — Pseudocode (6 marks)

book ← INPUT "Enter book title or ISBN"

found ← FALSE

```

FOR i ← 0 TO Books.length - 1
  IF Books[i].Title = book OR Books[i].ISBN = book THEN
    found ← TRUE
    IF Books[i].AvailableCopies > 0 THEN
      Books[i].AvailableCopies ← Books[i].AvailableCopies - 1
      OUTPUT "Book borrowed successfully"
      // Record loan
      newLoan.ISBN ← Books[i].ISBN
      newLoan.StudentID ← currentStudentID
      newLoan.BorrowDate ← TODAY
      newLoan.DueDate ← TODAY + 14
      ADD newLoan TO Loans
    ELSE
      OUTPUT "Book not available. Add to reservation list?"
      response ← INPUT
      IF response = "YES" THEN
        OUTPUT "Book reserved"
        // Add to reservations
      ENDIF
    ENDIF
  ENDIF
ENDIFOR
IF found = FALSE THEN
  OUTPUT "Book not found in library"
ENDIF

```

Mark allocation:

- Input book identifier (1)
- Search through book records (1)
- Check availability (1)
- Reduce available copies (1)
- Record loan with dates (1)
- Handle not available case (reservation) (1)
- Handle book not found (1)
- Output appropriate messages (1)

Max 6 marks. Accept equivalent logic with correct pseudocode syntax.

Part (c) — Security features (2 marks)

Any 2 valid features with reasons:

- Password/login: prevents unauthorised access to student accounts (1)
- Encryption: protects student personal data if database is breached (1)
- Access levels: librarians can edit, students can only view/borrow (1)
- Input validation: prevents SQL injection on search fields (1)
- Audit log: tracks who made changes and when (1)
- Backup: prevents data loss (1)

Part (d) — Testing (2 marks)

Any 2 valid test types with data and reason:

- Normal: valid book that is available → should successfully borrow (1)
- Boundary: book with AvailableCopies = 1 → should still work, then 0 (1)
- Boundary: book with AvailableCopies = 0 → should offer reservation (1)
- Erroneous: book title not in database → "not found" message (1)
- Erroneous: empty input → should handle gracefully (1)

(1 mark per test: type + data + reason)

EXAMINER NOTES — COMMON ERRORS TO AVOID

Part (a) common errors:

- Single table with all data (not normalised) → max 2 marks
- Missing foreign keys → deduct 1 mark
- No primary keys → deduct 1 mark

Part (b) common errors:

- Using = instead of ← for assignment
- Missing ENDIF or ENDFOR
- Not checking if book exists at all
- Not reducing available copies
- No error handling for unavailable books

Part (c) common errors:

- Just naming features without explaining WHY
- Generic answers: "use a password" without context
- Confusing security with validation

Part (d) common errors:

- Only giving one type of test data
- Test data without a reason
- Normal data only (no boundary/erroneous)

WEEK 14: MIXED PRACTICE & CONSOLIDATION

OVERVIEW:

Week 14 brings together everything from the targeted revision phase.

Students sit timed practice sections under exam conditions to build stamina, time management skills, and confidence. Immediate feedback helps them see progress and identify any final gaps before Mock Series 2. The mixed practice papers are deliberately focused on the topics the class has found most challenging, giving maximum impact in limited time.

LESSON 14.1 — MIXED PAPER 1 PRACTICE: 30-MINUTE TIMED SECTION

Topic: Timed Paper 1 Practice on Weak Areas

Duration: 40 min | Lesson Type: Timed Assessment | Syllabus Ref: 1.1–1.6

Learning Objective:

By the end of this lesson, students would have completed a 30-minute timed Paper 1 section under exam conditions and received immediate feedback on their performance.

Key Vocabulary:

- timed practice (计时练习 jìshí liànxí) — practice with a time limit
- exam conditions (考试条件 kǎoshì tiáojiàn) — same rules as real exam
- under pressure (在压力下 zài yālì xià) — working when time is limited
- stamina (耐力 nàilì) — ability to keep working for a long time
- immediate feedback (即时反馈 jíshí fǎnkuì) — getting results straight away

Lesson Structure (40 Minutes):

0–5 min | SET-UP AND INSTRUCTIONS

Teacher explains exam conditions:

- Work in silence
- No phones, no notes, no talking
- Show all working — method marks are crucial
- If stuck on a question, mark it and move on — come back if time
- 30 minutes for the paper, then 5 minutes checking

Distribute Worksheet 14A (Paper 1 Mixed Practice). Ensure all students have pen, pencil, ruler, calculator.

5–35 min | TIMED PRACTICE (30 MINUTES)

Students work in complete silence on the timed section.

Teacher invigilates:

- Walks around room (standard exam practice)
- Does not help with questions (builds independence)
- Notes any students who appear stuck for long periods
- Announces 15 minutes remaining, then 5 minutes remaining

Paper is 30 marks in 30 minutes — 1 minute per mark.

35–40 min | IMMEDIATE FEEDBACK

Teacher projects answer key. Students self-mark in a different colour:

- Tick correct answers
- Write correct working for any errors
- Total their mark

Quick poll: "Hands up if you got 20+/30" → celebrate.

"Hands up if you improved on this topic since last mock" → celebrate.

Teacher identifies any questions where >50% of class struggled → flag for last-minute revision.

EAL Support:

- All questions use clear, simple English
- Technical terms match those used throughout the course
- Students can ask for clarification of question wording ONLY (teacher can rephrase but not give answer)
- Answer key shown visually as well as verbally

Differentiation:

Emerging:

- May use notes/formula sheet during the first 10 minutes only
- Shorter paper option available (20 marks in 30 minutes)
- Teacher checks they are attempting all questions (not getting stuck)

Developing:

- Full timed paper under exam conditions
- Encouraged to attempt every question (no blanks)
- Self-mark with teacher support if needed

Secure:

- Full timed paper under strict exam conditions
- Extension: after self-marking, write examiner-style feedback for a peer's answer
- Target: 25+/30

Resources Needed:

- Worksheet 14A (Paper 1 Mixed Practice) — multiple copies
- Answer key for projection
- Pens, pencils, rulers, calculators
- Timer/stopwatch visible to class
- Formula reference cards (for Emerging only)

LESSON 14.2 — MIXED PAPER 2 PRACTICE: 30-MINUTE TIMED SECTION

Topic: Timed Paper 2 Practice on Weak Areas

Duration: 40 min | Lesson Type: Timed Assessment | Syllabus Ref: 2.1–2.3

Learning Objective:

By the end of this lesson, students would have completed a 30-minute timed Paper 2 section under exam conditions, self-marked their work, and identified any final gaps to address before Mock Series 2.

Key Vocabulary:

- mixed practice (混合练习 hùnhé liànxí) — practising different topics together
- consolidate (巩固 gǒnggù) — to make learning stronger
- identify gaps (找出差距 zhǎo chū chājù) — find what you still don't know
- final preparation (最后准备 zuìhòu zhǔnbèi) — getting ready at the end
- strict conditions (严格条件 yángé tiáojiàn) — following all the rules

Lesson Structure (40 Minutes):

0–5 min | SET-UP

Teacher reviews exam conditions (same as Lesson 14.1).

Reminders specific to Paper 2:

- Use correct pseudocode syntax (CAPS, ← not =, ENDIF present)
- Trace tables: one row per iteration, check your condition column
- SQL: remember JOIN needs a linking column
- Show all working in trace tables — every value, every row

Distribute Worksheet 14B (Paper 2 Mixed Practice).

5–35 min | TIMED PRACTICE (30 MINUTES)

Students work in silence.

Teacher invigilates and notes students who need support.

Time reminders at 15 minutes and 5 minutes.

35–40 min | IMMEDIATE FEEDBACK

Project answer key. Students self-mark.

Quick analysis:

- "Which question was hardest?" → class vote
- "Which topic do we need to revise one more time?" → teacher notes

Final encouragement:

- "You have done SO much revision. These mocks are your chance to show how far you've come."

Collect papers to review before Mock 4.

HOMEWORK: Revise your two weakest topics using your Personal Performance Tracker. Bring it to next lesson.

EAL Support:

- Same as Lesson 14.1
- Pseudocode syntax reminder on board during exam
- SQL keyword reference available (for all — this is standard exam preparation, not a crutch)

Differentiation:

Emerging:

- Notes allowed for first 10 minutes
- Shorter paper (20 marks) available
- Pseudocode syntax guide on desk

Developing:

- Full paper, exam conditions
- Trace table template provided (pre-drawn grid)

Secure:

- Full paper, strict conditions
- Extension: after self-marking, attempt a 6-mark pseudocode question from a different topic
- Target: 22+/30

Resources Needed:

- Worksheet 14B (Paper 2 Mixed Practice) — multiple copies
- Answer key for projection
- Pens, pencils, rulers
- Timer visible
- Syntax reference cards (Pseudocode + SQL)

--- WORKSHEET 14A ---

WORKSHEET 14A: MIXED PAPER 1 PRACTICE (30 marks, 30 minutes)

Name: _____ Date: _____ Start time: _____ End time: _____

Instructions:

- You have 30 minutes. There are 30 marks available.
- Answer ALL questions. Show ALL working — method marks are awarded.
- Calculators allowed. No notes, no phones.

Q1. Convert the following 8-bit binary number to denary. (1 mark)

1001 0110 = _____

Q2. Convert the denary number 189 to hexadecimal. Show your working. (2 marks)

Q3. Explain how ASCII is used to represent text in a computer. (2 marks)

Q4. An image has a resolution of 1280×720 pixels with 24-bit colour depth.

Calculate the uncompressed file size in MB. Show all working. (3 marks)

Q5. Explain why a JPEG file is typically smaller than an equivalent BMP file.

(3 marks)

Q6. Describe the purpose of the ALU and the Control Unit in the CPU. (4 marks)

Q7. Explain the fetch-execute cycle. Your answer should include the roles of

the MAR, MDR, PC and CIR. (5 marks)

Q8. Describe two differences between magnetic hard disk drives (HDD) and

solid-state drives (SSD). (2 marks)

Q9. Explain the purpose of an operating system. Give two functions. (2 marks)

Q10. Draw and label a diagram of the star network topology. Give one advantage

and one disadvantage of this topology. (3 marks)

Q11. Explain how a firewall helps to protect a computer network. (2 marks)

Q12. Describe one ethical concern related to the increasing use of artificial

intelligence in the workplace. (1 mark)

TOTAL: _____ / 30

--- ANSWER KEY: WORKSHEET 14A ---

ANSWER KEY: WORKSHEET 14A — MIXED PAPER 1 PRACTICE (30 marks)

Q1. Binary to denary (1 mark)

$$128 + 16 + 4 + 2 = 150$$

Answer: 150

Q2. 189 to hex (2 marks)

$$189 \div 16 = 11 \text{ remainder } 13 \text{ (D)}$$

$$11 = \text{B}$$

Answer: BD (1 mark method, 1 mark answer)

Q3. ASCII (2 marks)

— Each character is represented by a unique binary number (1)

— Standard ASCII uses 7 bits (128 characters), extended uses 8 bits (1)

— Stored as binary in memory (1)

Max 2 marks

Q4. Image file size (3 marks)

$$1280 \times 720 \times 24 = 22,118,400 \text{ bits}$$

$$\div 8 = 2,764,800 \text{ bytes}$$

$\div 1024 = 2700 \text{ KB}$

$\div 1024 = 2.64 \text{ MB}$ (accept 2.6 MB)

(1 mark formula, 1 mark calculation, 1 mark MB conversion)

Q5. JPEG vs BMP (3 marks)

- JPEG uses lossy compression, BMP is typically uncompressed (1)
- JPEG permanently removes data that human eye is less sensitive to (1)
- BMP stores every pixel individually, no compression (1)
- This makes JPEG significantly smaller, especially for photographs (1)

Max 3 marks

Q6. ALU and CU (4 marks)

ALU:

- Performs arithmetic operations (addition, subtraction) (1)
- Performs logical comparisons (AND, OR, NOT, greater than, equals) (1)

Control Unit:

- Manages and coordinates all CPU operations (1)
- Sends control signals to other components (1)
- Decodes instructions (1)
- Manages fetch-execute cycle (1)

Max 4 marks (2 per component)

Q7. Fetch-execute cycle (5 marks)

- PC holds address of next instruction to be fetched (1)
- Address copied from PC to MAR (1)
- Address placed on address bus and sent to memory (1)
- Instruction at that address copied to MDR via data bus (1)
- Instruction copied from MDR to CIR (1)
- Instruction decoded by Control Unit (1)
- Instruction executed by ALU (1)
- PC incremented to point to next instruction (1)

Max 5 marks

Q8. HDD vs SSD (2 marks)

Any 2 valid differences:

- HDD has moving parts (spinning disk), SSD has no moving parts (1)
- SSD faster read/write speeds than HDD (1)
- SSD uses less power than HDD (1)
- SSD more expensive per GB (1)
- SSD lighter/more shock-resistant (1)

Q9. Operating system (2 marks)

— Purpose: manages computer hardware and software resources (1)

Any 2 functions: memory management, processor management/scheduling,
file management, security, device management, user interface (1 mark total)

Q10. Star topology (3 marks)

— Diagram: central switch/hub with all devices connected directly (1)

— Advantage: fault in one cable doesn't affect others / easy to add
devices / easy to diagnose (1)

— Disadvantage: if central device fails whole network fails /
expensive cabling (1)

Q11. Firewall (2 marks)

— Monitors incoming and outgoing network traffic (1)

— Blocks unauthorised access based on security rules / filters packets
by IP address or port number (1)

Q12. AI ethical concern (1 mark)

Any valid concern:

— Job displacement / unemployment

— Bias in training data leading to unfair decisions

— Privacy concerns from data collection

— Lack of accountability when AI makes mistakes

— Over-reliance on technology

(1 mark for any valid concern with brief explanation)

--- WORKSHEET 14B ---

WORKSHEET 14B: MIXED PAPER 2 PRACTICE (30 marks, 30 minutes)

Name: _____ Date: _____ Start time: _____ End time: _____

Instructions:

- You have 30 minutes. There are 30 marks available.
- Use correct pseudocode syntax (CAPS, $\leftarrow$ not =, ENDIF, ENDFOR, ENDWHILE).
- Show ALL working in trace tables.
- No notes, no phones.

Q1. What is the output of this pseudocode? (2 marks)

total $\leftarrow$ 0

FOR i $\leftarrow$ 1 TO 4

total $\leftarrow$ total + (i $\times$ i)

ENDFOR

OUTPUT total

Q2. Trace the following pseudocode: (5 marks)

```
num ← 12
count ← 0
WHILE num > 1
  IF num MOD 2 = 0 THEN
    num ← num DIV 2
  ELSE
    num ← num × 3 + 1
  ENDIF
  count ← count + 1
ENDWHILE
OUTPUT count
```

Complete this trace table:

num	count	num MOD 2 = 0?	Action	num > 1?
12	0			

OUTPUT: _____

Q3. Write pseudocode that inputs 10 numbers and calculates how many are positive and how many are negative. Output both counts. (4 marks)

Q4. Tables: Student(StudentID, Name, Age, Class) and

Result(ResultID, StudentID, Subject, Mark)

a) Write an SQL query to find the names of all students in Class 10A.

(2 marks)

b) Write an SQL query to display each student's name and their marks in all subjects. (2 marks)

c) Write an SQL query to find the average mark for each subject.

(2 marks)

Q5. Draw a logic circuit for $Q = (\text{NOT } A \text{ OR } B) \text{ AND } C$. Label all inputs, gates, and the output. (3 marks)

Q6. Complete the truth table for $Q = (\text{NOT } A \text{ OR } B) \text{ AND } C$: (4 marks)

A	B	C	NOT A	NOT A OR B	Q
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			

Q7. A program needs to validate that a user has entered a valid email address (must contain @ and .). Write pseudocode for this validation. (4 marks)

Q8. Write a Python function called count_vowels that takes a string and returns the number of vowels (a, e, i, o, u) in the string. (4 marks)

Q9. Explain why it is important to use normalisation when designing a database. (2 marks)

Q10. Simplify this Boolean expression using Boolean algebra laws.

Show each step. (2 marks)

$X = (A \text{ AND } B) \text{ OR } (A \text{ AND NOT } B)$

--- ANSWER KEY: WORKSHEET 14B ---

ANSWER KEY: WORKSHEET 14B — MIXED PAPER 2 PRACTICE (30 marks)

Q1. Pseudocode output (2 marks)

total = 0 + 1 + 4 + 9 + 16 = 30

OUTPUT: 30 (2 marks)

(1 mark for method showing working, 1 mark correct answer)

Q2. Trace table (5 marks)

num	count	num MOD 2 = 0?	Action	num > 1?
12	0	Yes (0)	num ← 6	True
6	1	Yes (0)	num ← 3	True
3	2	No (1)	num ← 10	True
10	3	Yes (0)	num ← 5	True
5	4	No (1)	num ← 16	True
16	5	Yes (0)	num ← 8	True
8	6	Yes (0)	num ← 4	True
4	7	Yes (0)	num ← 2	True
2	8	Yes (0)	num ← 1	False

OUTPUT: 9

(This is the Collatz conjecture sequence — not required to name it)

(0.5 marks per correct row, max 4 + 1 mark output)

Deduct 0.5 for each incorrect row.

Q3. Positive/negative count (4 marks)

pos ← 0

neg ← 0

FOR i ← 1 TO 10

num ← INPUT

```

IF num > 0 THEN
    pos ← pos + 1
ELSE IF num < 0 THEN
    neg ← neg + 1
ENDIF
ENDFOR
OUTPUT pos, neg
(1 mark variables, 1 mark loop, 1 mark IF structure, 1 mark output)

```

Q4. SQL queries (6 marks)

- a) SELECT Name FROM Student WHERE Class = '10A'; (2 marks)
- b) SELECT Student.Name, Result.Subject, Result.Mark
FROM Student INNER JOIN Result
ON Student.StudentID = Result.StudentID; (2 marks)
- c) SELECT Subject, AVG(Mark) AS AverageMark
FROM Result
GROUP BY Subject; (2 marks)

Q5. Logic circuit (3 marks)

- A goes through NOT gate (1)
- NOT A and B go into OR gate (1)
- Output of OR and C go into AND gate → Q (1)

Award marks for correct structure even if drawing is rough.

Q6. Truth table (4 marks)

A	B	C	NOT A	NOT A OR B	Q
0	0	0	1	1	0
0	0	1	1	1	1
0	1	0	1	1	0
0	1	1	1	1	1
1	0	0	0	0	0
1	0	1	0	0	0
1	1	0	0	1	0
1	1	1	0	1	1

(0.5 marks per correct row)

Q7. Email validation (4 marks)

```

email ← INPUT
has_at ← FALSE
has_dot ← FALSE

```

```
FOR i ← 0 TO LENGTH(email) - 1
```

```
  IF email[i] = '@' THEN
```

```
    has_at ← TRUE
```

```
  ENDIF
```

```
  IF email[i] = '.' THEN
```

```
    has_dot ← TRUE
```

```
  ENDIF
```

```
ENDFOR
```

```
IF has_at = TRUE AND has_dot = TRUE THEN
```

```
  OUTPUT "Valid email"
```

```
ELSE
```

```
  OUTPUT "Invalid email"
```

```
ENDIF
```

(1 mark input, 1 mark loop through characters, 1 mark checking @ and .,
1 mark output)

Accept alternative approaches (e.g. using string functions if specified).

Q8. Python function (4 marks)

```
def count_vowels(text):
```

```
    vowels = "aeiouAEIOU"
```

```
    count = 0
```

```
    for char in text:
```

```
        if char in vowels:
```

```
            count = count + 1
```

```
    return count
```

(1 mark def/name/parameter, 1 mark vowel definition, 1 mark loop,
1 mark return)

Accept case-sensitive or case-insensitive versions.

Q9. Normalisation (2 marks)

— Removes data redundancy / duplication (1)

— Prevents update anomalies / maintains data consistency / organises
data efficiently into related tables (1)

Q10. Boolean simplification (2 marks)

$X = (A \text{ AND } B) \text{ OR } (A \text{ AND NOT } B)$

$X = A \text{ AND } (B \text{ OR NOT } B) \leftarrow \text{distributive law (1)}$

$X = A \text{ AND } 1 \leftarrow \text{complement } (B \text{ OR NOT } B = 1)$

$X = A \leftarrow \text{identity (1)}$

(1 mark per correct step with law named)

UNIT 5 OVERVIEW

Mock Series 2

Unit 5 spans Weeks 15–16 and comprises the final two full mock exams before

the IGCSE examinations. These are high-stakes assessments designed to:

1. Give students a final experience of full exam conditions

2. Provide accurate predicted grades

47. Identify any remaining topics for last-minute intervention

4. Build exam confidence through successful performance

Mock 4 (Week 15): Full Paper 1 — Theory of Computer Science

Mock 5 (Week 16): Full Paper 2 — Problem Solving & Programming

Both papers are 75 marks, 1 hour 45 minutes, under full exam conditions.

WEEK 15: MOCK EXAM 4 — PAPER 1

OVERVIEW:

Mock 4 is a full Paper 1 examination (75 marks, 1 hour 45 minutes). It covers the complete Paper 1 syllabus and is designed to be of similar difficulty to the actual CIE IGCSE 0478 Paper 1. The paper is DIFFERENT from Mock 2 (which was also Paper 1) — all questions are original.

Lesson 15.1: Full mock exam under exam conditions

Lesson 15.2: Review, self-marking, progress comparison

LESSON 15.1 — MOCK EXAM 4: FULL PAPER 1 (1 hour 45 minutes)

Topic: Full Paper 1 Mock Examination

Duration: 1h 45min | Lesson Type: Formal Examination | Syllabus Ref: 1.1–1.6

Learning Objective:

To complete a full Paper 1 examination under exam conditions, demonstrating knowledge and understanding of all Paper 1 topics.

Lesson Structure (105 Minutes — note: this may span a double lesson):

0–5 min | SET-UP

Teacher ensures exam conditions:

- Desks spaced apart (or alternative seating as per school policy)
- No materials on desks except: pen (black/blue), pencil, ruler, eraser, calculator
- No phones, no watches with data storage, no notes
- Clock visible to all students
- Paper seal broken together under teacher supervision

Teacher reads instructions:

"You have 1 hour 45 minutes. There are 75 marks. Answer ALL questions. Show all working. Write in black or blue pen. No calculators for binary/hex conversions — show your working."

5–100 min | EXAMINATION

Students work in complete silence.

Teacher invigilates according to exam regulations.

Time warnings: 60 min remaining, 30 min remaining, 10 min remaining, 5 min remaining, time up.

100–105 min | COLLECTION

All papers collected in order. Students remain seated until all papers are accounted for. Students dismissed.

Teacher marks papers (or arranges for external marking) before Lesson 15.2.

EAL Support:

- EAL students may request clarification of question wording (teacher may rephrase but NOT explain content)
- EAL students have same time allocation — no extra time unless specified in access arrangements
- Technical terms are standard IGCSE vocabulary that all students have encountered throughout the course

Differentiation:

Emerging:

- Seated near front for teacher observation
- Encouraged to attempt every question (even if unsure)
- Reminded about method marks for showing working

Developing:

- Standard exam conditions
- Target: achieve Grade C equivalent (44/75)

Secure:

- Standard exam conditions

— Target: achieve Grade A/A* equivalent (60+/75)

Resources Needed:

- Mock Exam 4 Paper 1 papers (sufficient copies for all students)
- Answer booklets or lined paper
- Exam attendance register
- Clock/stopwatch
- "Silence — Exam in Progress" signs for doors

LESSON 15.2 — MOCK 4 REVIEW: SELF-MARKING AND PROGRESS TRACKING

Topic: Exam Review, Self-Assessment & Progress Analysis

Duration: 40 min | Lesson Type: Review Workshop | Syllabus Ref: 1.1–1.6

Learning Objective:

By the end of this lesson, students will have self-marked their Mock 4, compared their performance across all Paper 1 mocks, identified their progress trajectory, and set final targets.

Key Vocabulary:

- self-mark (自评 zì píng) — marking your own work
- progress trajectory (进步轨迹 jìnbù guǐjì) — path of improvement over time
- trajectory (趋势 qūshì) — direction of change
- celebrate (庆祝 qìngzhù) — recognise achievement
- remaining (剩余的 shèngyú de) — still left to do

Lesson Structure (40 Minutes):

0–3 min | DO NOW

Students receive their marked Mock 4 paper (if teacher-marked) or their unmarked paper with answer key (if self-marking).

They also receive the "Mock Progress Tracker" (Worksheet 15A).

Write Mock 4 mark in the space provided.

3–12 min | SELF-MARKING (if not teacher-marked)

OR: Review of teacher-marked papers.

Students go through the mark scheme question by question:

- Green pen for correct answers
- Red pen for corrections
- Note marks gained and lost per question

Teacher circulates to verify self-marking is accurate.

12–25 min | PROGRESS ANALYSIS

Students complete the Mock Progress Tracker (Worksheet 15A):

- Mock 1 Paper 1 mark → Mock 2 Paper 1 mark → Mock 4 Paper 1 mark
- Plot on graph (visual progress line)

- Calculate improvement/drop between each mock
- Identify: "What helped me improve?" or "What caused the drop?"

Teacher reveals class statistics:

- Class average, range, highest
- Grade distribution histogram
- Comparison with Mock 2 (same paper type)

CELEBRATION: Students who improved receive verbal recognition.

"Most improved" and "Highest score" acknowledged (with consent).

25–35 min | INDIVIDUAL TARGET-SETTING

Students write:

- My Mock 4 grade: _____
- My target for real exam: _____
- To reach my target I need to: (3 specific actions)
- The topic I will revise TONIGHT is: _____

Teacher conducts brief individual conferences (30 sec each):

- "Your progress is... / Your priority is... / Keep doing..."
- Focus on students who dropped or stayed the same

35–40 min | PLENARY + HOMEWORK

"Three words to describe how I feel about Paper 1 now" — students write on sticky notes, post on "Confidence Wall."

HOMEWORK: Revise your weakest Paper 1 topic for 30 minutes. Use your notes, flashcards, or online resources. Bring one question to next lesson.

EAL Support:

- Mark scheme available with Chinese topic labels
- Progress tracker has visual graph (less language-dependent)
- Teacher conferences use simple, encouraging language
- Peer support: bilingual students can help explain mark scheme

Differentiation:

Emerging:

- Teacher marks paper in advance (no self-marking)
- Progress tracker pre-filled with marks
- Conference focuses on ONE priority topic only
- Simplified target-setting (one target, not three)

Developing:

- Self-mark with teacher support
- Full progress tracker

- Two specific targets set

Secure:

- Self-mark independently
- Full tracker + extension analysis (which question types still lose marks? — application vs recall?)
- Support peers during progress analysis

Resources Needed:

- Marked or unmarked Mock 4 papers
- Full mark scheme (projected or printed)
- Worksheet 15A — Mock Progress Tracker
- Green and red pens
- Graph paper for progress line
- Sticky notes for Confidence Wall

MOCK EXAM 4: IGCSE COMPUTER SCIENCE (0478) PAPER 1

THEORY OF COMPUTER SCIENCE

Time: 1 hour 45 minutes Maximum Marks: 75

INSTRUCTIONS:

- Answer ALL questions.
- Use black or blue pen. You may use pencil for diagrams.
- Show ALL your working. Marks are awarded for correct methods even if your final answer is wrong.
- No calculators for binary or hexadecimal conversions.
- Calculators ARE allowed for file size calculations.

MARK ALLOCATION BY TOPIC (for information only):

Data Representation 16 marks

Data Transmission 8 marks

Hardware 16 marks

Software 8 marks

The Internet & Its Uses 12 marks

Automated Systems & Emerging Tech .. 15 marks

QUESTION 1 [6 marks]

Data Representation — Number Systems

- Convert the 8-bit binary number 1011 0100 to denary. (1)
- Convert the denary number 237 to hexadecimal. Show your working. (2)
- Convert the hexadecimal number 3E8 to denary. Show your working. (2)
- Explain why hexadecimal is often used when representing binary data in computing, rather than displaying the full binary number. (1)

QUESTION 2 [7 marks]

Data Representation — Text, Sound, and Images

- (a) Describe how a text character is represented using the Unicode system. (2)
- (b) A sound recording is made with a sampling rate of 32 kHz and a sample resolution of 16 bits. The recording lasts for 2 minutes.
Calculate the file size of the uncompressed audio file in megabytes.
Show all your working. (4)
- (c) Explain why increasing the sampling rate improves the quality of a digital sound recording. (1)

QUESTION 3 [5 marks]

Data Representation — Compression

- (a) A company stores large numbers of photographic images. Explain why the company might choose to use lossy compression rather than lossless compression for these images. (2)
- (b) The following string of characters is to be compressed using run-length encoding (RLE):
A A A A B C C C C C D D A A B B B
Show the result after RLE compression. Calculate the compression ratio.
(3)

QUESTION 4 [4 marks]

Data Transmission — Error Detection

- (a) Explain how a parity bit is used to detect errors in data transmission. (2)
- (b) A system uses odd parity. The following byte of data is received:
(the last bit is the parity bit)
Has an error occurred? Explain your answer. (2)

QUESTION 5 [4 marks]

Data Transmission — Check Digits

A product code uses a weighted check digit system. The code is a 4-digit number D1 D2 D3 D4, where D4 is the check digit.

The check digit is calculated as:

$$\text{Check digit} = (D1 \times 3 + D2 \times 1 + D3 \times 3) \text{ MOD } 10$$

- (a) Calculate the check digit for the product code 5 7 2 _ . (2)
- (b) A customer enters the product code 5 7 2 8. Explain how the system would verify this code and what would happen. (2)

QUESTION 6 [8 marks]

Hardware — The CPU and Fetch-Execute Cycle

(a) State the purpose of each of the following registers in the CPU:

- (i) PC (Program Counter) (1)
- (ii) MAR (Memory Address Register) (1)
- (iii) MDR (Memory Data Register) (1)
- (iv) CIR (Current Instruction Register) (1)

(b) Describe the fetch-execute cycle. Your answer should explain the role of the registers in part (a) and the buses connecting the CPU to memory. (4)

QUESTION 7 [5 marks]

Hardware — Memory

(a) Explain the difference between RAM and ROM. Your answer should refer to volatility and typical use in a computer system. (3)

(b) Explain why adding more RAM to a computer can improve its performance when running multiple programs at the same time. (2)

QUESTION 8 [3 marks]

Hardware — Storage Devices

A student needs to buy storage for their computer. They are choosing between a magnetic hard disk drive (HDD) and a solid-state drive (SSD).

(a) Give two advantages of using an SSD compared to an HDD. (2)

(b) Give one disadvantage of using an SSD compared to an HDD. (1)

QUESTION 9 [5 marks]

Software — Operating Systems and Utility Software

(a) Describe two functions of an operating system. (2)

(b) Explain the purpose of the following utility programs:

- (i) Disk defragmenter (2)
- (ii) Backup software (1)

QUESTION 10 [5 marks]

Software — Types of Software

(a) Explain the difference between system software and application software.

Give one example of each. (2)

(b) Describe what is meant by open-source software. Give one advantage and one disadvantage of using open-source software. (3)

QUESTION 11 [4 marks]

The Internet — Network Hardware and Protocols

(a) Describe the purpose of a router in a computer network. (2)

(b) Explain how the TCP protocol ensures reliable delivery of data across

the internet. (2)

QUESTION 12 [4 marks]

The Internet — Network Topologies and Models

- (a) Draw and describe the bus network topology. Give one advantage and one disadvantage of using a bus topology. (3)
- (b) Describe one difference between the client-server model and the peer-to-peer model of networking. (1)

QUESTION 13 [4 marks]

The Internet — IP Addressing and the Domain Name System

- (a) Explain why IP addresses are necessary for data to be sent across the internet. (2)
- (b) Describe the purpose of the Domain Name System (DNS). (2)

QUESTION 14 [6 marks]

Internet Security

- (a) Describe what is meant by a denial-of-service (DoS) attack. (2)
- (b) Explain two methods that can be used to prevent malware from infecting a computer system. (4)

QUESTION 15 [5 marks]

Automated Systems — Robotics and Artificial Intelligence

A factory uses robots on an assembly line to build cars.

- (a) Describe two sensors that might be used by the robots and explain what each sensor would measure. (2)
- (b) Explain two advantages of using robots rather than human workers on an assembly line. (2)
- (c) Describe one disadvantage of replacing human workers with robots. (1)

QUESTION 16 [5 marks]

Emerging Technologies — The Internet of Things and Cloud Computing

- (a) Describe what is meant by the Internet of Things (IoT). Give one example of an IoT device in a smart home. (2)
- (b) Explain what is meant by cloud computing. Give two advantages for a business of storing data in the cloud rather than on local servers. (3)

END OF PAPER

MARK SCHEME — MOCK EXAM 4 PAPER 1

QUESTION 1 [6 marks]

- (a) $1011\ 0100 = 128 + 32 + 16 + 4 = 180$ (1)
- (b) $237 \div 16 = 14$ remainder 13 (D) (1)

14 = E

Answer: ED (1)

Method mark for showing division by 16. Answer mark for correct hex.

(c) $(3 \times 256) + (14 \times 16) + 8$ (1)

= $768 + 224 + 8 = 1000$ (1)

Method mark for showing positional values. Answer mark for correct denary.

(d) — Hexadecimal is more compact/shorter than binary (1)

— Easier for humans to read and remember (1)

— One hex digit represents exactly 4 binary digits (1)

Max 1 mark

QUESTION 2 [7 marks]

(a) — Each character is assigned a unique binary code/number (1)

— Unicode uses 16 bits (or more), allowing many more characters than ASCII / supports characters from many languages (1)

— Stored as binary in computer memory (1)

Max 2 marks

(b) 32 kHz = 32,000 Hz (1)

2 minutes = 120 seconds (1)

File size = $32,000 \times 16 \times 120 = 61,440,000$ bits (1)

$\div 8 = 7,680,000$ bytes

$\div 1024 = 7500$ KB

$\div 1024 = 7.32$ MB (accept 7.3 MB) (1)

Award 1 mark for correct formula with substitution, 1 for bits,

1 for MB. Accept calculation showing steps clearly.

(c) — Higher sampling rate captures more samples per second (1)

— More accurate representation of the original analogue sound wave (1)

— Less detail is lost between samples (1)

Max 1 mark

QUESTION 3 [5 marks]

(a) — Lossy compression permanently removes some data (1)

— For photographs, some data loss is not noticeable to the human eye (1)

— Much smaller file sizes than lossless (1)

— Faster to upload/download / less storage space needed (1)

Max 2 marks

(b) A4 B1 C5 D2 A2 B3 (or equivalent clear notation) (1)

Original: 17 characters

Compressed: 12 items/values (1)

Compression ratio = 17:12 or 1.42:1 (accept equivalent) (1)

QUESTION 4 [4 marks]

(a) — An extra bit added to each byte of data (1)

— Parity bit set to make total number of 1s either odd (odd parity)
or even (even parity) (1)

— If received parity doesn't match, an error is detected (1)

Max 2 marks

(b) Byte (without parity): 1 0 1 0 1 1 0 — count of 1s = 4 (even) (1)

System uses ODD parity but 4 is even, so error detected (1)

OR: Count of 1s including parity bit = 4 (even), but odd parity
expected, so error has occurred.

QUESTION 5 [4 marks]

(a) $(5 \times 3 + 7 \times 1 + 2 \times 3) = 15 + 7 + 6 = 28$ (1)

$28 \text{ MOD } 10 = 8$

Check digit = 8 (1)

(b) System recalculates: $(5 \times 3 + 7 \times 1 + 2 \times 3) \text{ MOD } 10 = 8$ (1)

Entered check digit is 8, matches calculation → code accepted (1)

QUESTION 6 [8 marks]

(a) (i) PC — holds address of next instruction to be fetched (1)

(ii) MAR — holds memory address being read from or written to (1)

(iii) MDR — holds data/instruction being transferred to/from memory (1)

(iv) CIR — holds current instruction being decoded and executed (1)

(b) — PC content copied to MAR (1)

— Address placed on address bus, sent to memory (1)

— Instruction at that address placed in MDR via data bus (1)

— Instruction copied from MDR to CIR (1)

— PC incremented (1)

— Instruction decoded by CU, executed by ALU (1)

— Process repeats continuously (1)

Max 4 marks

QUESTION 7 [5 marks]

(a) RAM:

— Volatile: loses contents when power removed (1)

— Used for currently running programs and data (1)

ROM:

- Non-volatile: retains contents when power removed (1)
- Used for boot/start-up instructions / BIOS (1)

Max 3 marks (must mention volatility for full marks)

- (b) — More RAM means more programs can run simultaneously without using virtual memory (1)
- Virtual memory (HDD/SSD used as RAM) is much slower than RAM, so avoiding it improves performance (1)
 - Less swapping/data transfer between RAM and secondary storage (1)

Max 2 marks

QUESTION 8 [3 marks]

(a) Any 2 from:

- Faster read/write speeds (1)
- No moving parts, so more reliable / less likely to fail (1)
- Uses less power (1)
- Lighter / more portable (1)
- Silent operation (1)
- Less heat produced (1)

(b) Any 1 from:

- More expensive per GB (1)
- Smaller maximum capacities available (1)
- Data harder to recover if failure occurs (1)

QUESTION 9 [5 marks]

(a) Any 2 functions with description:

- Memory management: allocates RAM to programs, manages virtual memory (1)
- Processor management: schedules tasks, allocates CPU time (1)
- File management: organises storage, handles file creation/deletion (1)
- Security: manages user accounts, passwords, access rights (1)
- Device management: controls peripheral devices via drivers (1)
- User interface: provides GUI or CLI for user interaction (1)

Max 2 marks (1 per function)

- (b) (i) — Defragmentation rearranges files on disk so they are stored in contiguous blocks (1)
- Reduces read/write head movement, speeding up file access (1)
- (ii) — Creates copies of data that can be restored if original data is lost or corrupted (1)

QUESTION 10 [5 marks]

(a) — System software: manages computer hardware and provides platform for applications / controls and coordinates system resources (1)

Example: operating system, utility software, device drivers (1)

— Application software: programs that perform specific tasks for the user (1)

Example: word processor, spreadsheet, web browser, game (1)

Max 2 marks (1 for difference, 1 for examples)

(b) — Open-source: source code is freely available and can be modified and redistributed by anyone (1)

— Advantage: free to use / large community support / can be customised / transparent security (1)

— Disadvantage: may lack official support / less user-friendly / compatibility issues / no single organisation responsible (1)

QUESTION 11 [4 marks]

(a) — Forwards data packets between networks (1)

— Uses IP addresses to determine best path for data (1)

— Connects LAN to internet / different networks together (1)

Max 2 marks

(b) — TCP breaks data into packets and reassembles them at destination (1)

— Checks for missing packets and requests retransmission (1)

— Verifies data integrity using checksums (1)

— Manages flow control / ensures packets arrive in correct order (1)

Max 2 marks

QUESTION 12 [4 marks]

(a) — Diagram: single central cable with all devices connected (1)

— All data travels along the central bus in both directions (1)

— Advantage: cheap / easy to install / less cable than star (1)

— Disadvantage: if main cable fails whole network fails / difficult to diagnose faults / collisions slow down network (1)

Max 3 marks (1 diagram, 1 advantage, 1 disadvantage)

(b) — Client-server: central server provides services to clients (1)

— Peer-to-peer: all computers equal, share resources directly (1)

Max 1 mark

QUESTION 13 [4 marks]

(a) — Every device on the internet has a unique IP address (1)

— Routers use destination IP address to route packets to correct device (1)

- Without IP addresses, data would not know where to go (1)

Max 2 marks

(b) — DNS translates domain names into IP addresses (1)

- Users can use memorable names (e.g. www.google.com) instead of remembering IP addresses (1)
- DNS servers store the mappings between names and addresses (1)

Max 2 marks

QUESTION 14 [6 marks]

(a) — Overwhelming a server or network with excessive traffic/requests (1)

- Makes service unavailable to legitimate users (1)
- Uses multiple compromised computers (DDoS) or single source (1)

Max 2 marks

(b) Any 2 methods:

- Install and regularly update antivirus software (1 + reason 1)
Reason: detects and removes malware before it can cause damage
- Use a firewall to block unauthorised access (1 + reason 1)
- Keep operating system and software updated with security patches (1 + 1)
- Don't open email attachments from unknown sources (1 + 1)
- Only download software from trusted websites (1 + 1)
- Use strong passwords and access controls (1 + 1)

Max 4 marks (1 per method + 1 per reason)

QUESTION 15 [5 marks]

(a) Any 2 sensors with measurement:

- Proximity sensor: measures distance to nearby objects (1)
- Temperature sensor: measures heat to detect overheating (1)
- Pressure sensor: measures force applied during assembly (1)
- Light sensor: detects presence/absence of parts (1)
- Position/orientation sensor: ensures parts correctly aligned (1)

Max 2 marks

(b) Any 2 advantages:

- Robots work 24/7 without breaks / faster production (1)
- More consistent quality / fewer errors (1)
- Can work in dangerous environments / handle hazardous materials (1)
- Don't need wages, holidays, or sick leave / lower long-term costs (1)

Max 2 marks

(c) Any 1 disadvantage:

- High initial cost of purchase and installation (1)
- Workers lose jobs / unemployment (1)
- Require skilled technicians to maintain / expensive repairs (1)
- Less flexible than humans / can't adapt to unexpected situations (1)

Max 1 mark

QUESTION 16 [5 marks]

(a) — IoT: everyday physical devices connected to the internet, collecting and exchanging data (1)

- Example: smart thermostat, smart light bulb, smart fridge, wearable fitness tracker, smart doorbell (1)

Max 2 marks

(b) — Cloud computing: delivering computing services (storage, processing, software) over the internet (1)

- Advantage 1: accessible from anywhere with internet connection (1)
- Advantage 2: scalable / pay for what you use / no need to maintain own hardware / automatic backups / reduced IT costs (1)

Max 3 marks

GRADE BOUNDARIES — MOCK EXAM 4 PAPER 1

A*: 69–75 (92–100%)

A: 61–68 (81–91%)

B: 53–60 (71–80%)

C: 45–52 (60–70%)

D: 37–44 (49–59%)

E: 29–36 (39–48%)

F: 21–28 (28–38%)

G: 13–20 (17–27%)

U: 0–12 (0–16%)

--- WORKSHEET 15A ---

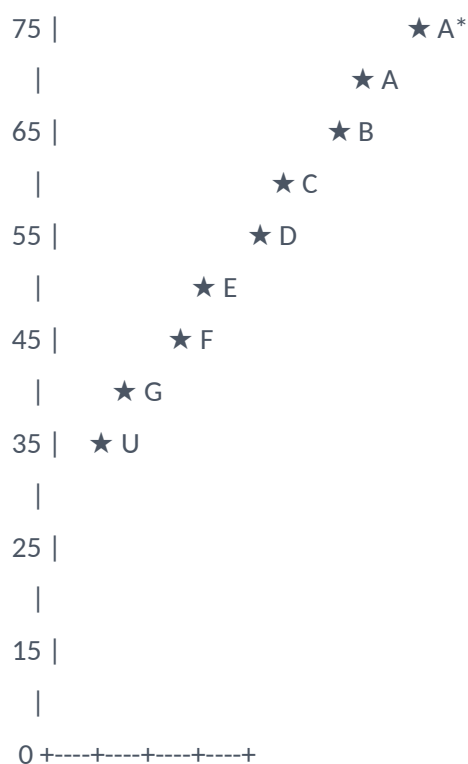
WORKSHEET 15A: MOCK PROGRESS TRACKER (Paper 1 Journey)

Name: _____ Date: _____

MY PAPER 1 MOCK RESULTS OVER TIME

Mock	Date	Mark /75	%	Grade	Topics I Lost Marks On
M1					
M2					
M4					

Graph: Plot your marks on the line below



● M1: ____/75 ● M2: ____/75 ● M4: ____/75

MY PROGRESS ANALYSIS

Improvement M1→M2: _____ marks (up/down/same)

Improvement M2→M4: _____ marks (up/down/same)

Total improvement M1→M4: _____ marks

My trajectory is: ☐ Improving steadily ☐ Improving then dropped

☐ Dropped then improved ☐ Stayed similar

What helped me improve? (or what caused a drop?)

MY FINAL TARGETS

My Mock 4 grade: _____ My target for real exam: _____

Three specific actions to reach my target:

48. _____

49. _____

50. _____

The topic I will revise TONIGHT:

CONFIDENCE SCALE (circle):

I feel confident about Paper 1: 1 — 2 — 3 — 4 — 5

(1 = not at all, 5 = very confident)

--- ANSWER KEY: WORKSHEET 15A ---

ANSWER KEY: WORKSHEET 15A — MOCK PROGRESS TRACKER

This is a reflective worksheet — no "right" answers. Mark for completion.

MARKING CRITERIA (10 marks):

- Mock results table completed accurately (3 marks)
 - All 3 mocks filled in with correct marks (1)
 - Percentages calculated correctly: $(\text{mark} \div 75) \times 100$ (1)
 - Grades correct from boundaries (1)
- Graph plotted correctly (2 marks)
 - All 3 points in approximately correct position (1)
 - Points labelled or clear which is which (1)
- Progress analysis completed meaningfully (2 marks)
 - Differences calculated correctly (1)
 - Written reflection has specific detail (1)
- Targets are SMART (Specific, Measurable, Achievable, Relevant, Time-bound) (2 marks — 1 per quality target, max 2)
- Confidence scale completed (1 mark)

TEACHER NOTES:

- This tracker is designed for students to SEE their progress visually.
- Celebrate ALL improvement, even 1 mark.
- Students who dropped between mocks need a supportive conversation — focus on what caused the drop (often topic knowledge, not ability).
- Use confidence scale data to identify students who need emotional support before the real exam.

WEEK 16: MOCK EXAM 5 — PAPER 2

OVERVIEW:

Mock 5 is a full Paper 2 examination (75 marks, 1 hour 45 minutes). It covers the complete Paper 2 syllabus including algorithms, programming, databases, Boolean logic, and the 15-mark scenario question. The paper is DIFFERENT from Mock 3 (which was also Paper 2) — all questions are original.

Lesson 16.1: Full mock exam under exam conditions

Lesson 16.2: Review, self-marking, individual conferences, final targets

LESSON 16.1 — MOCK EXAM 5: FULL PAPER 2 (1 hour 45 minutes)

Topic: Full Paper 2 Mock Examination

Duration: 1h 45min | Lesson Type: Formal Examination | Syllabus Ref: 2.1–2.3

Learning Objective:

To complete a full Paper 2 examination under exam conditions, demonstrating problem-solving and programming skills.

Lesson Structure (105 Minutes):

0–5 min | SET-UP

Same exam conditions as Mock 4 (Week 15).

Additional instructions for Paper 2:

- Use correct pseudocode syntax. Keywords should be in CAPS.
- Use $\leftarrow$ for assignment (NOT =).
- Show ALL working in trace tables — every row, every value.
- For SQL questions, write complete queries with correct keywords.
- For the 15-mark question, PLAN before you write. Use the planning space provided.

5–100 min | EXAMINATION

Full exam conditions. Teacher invigilates.

Time warnings at standard intervals.

100–105 min | COLLECTION

Papers collected. Students dismissed.

EAL Support:

- Same as Mock 4
- Pseudocode syntax rules are standard — all students have practised these extensively

Differentiation:

Emerging:

- Encouraged to attempt every question
- Reminded to show working (method marks available even if final answer is wrong)
- Focus on getting marks on recall questions (definitions, naming)

Developing:

- Standard exam conditions
- Target: achieve Grade C equivalent (44/75)

Secure:

- Standard exam conditions
- Target: achieve Grade A/A* equivalent (60+/75)
- Should complete all questions including 15-mark scenario

Resources Needed:

- Mock Exam 5 Paper 2 papers — multiple copies
- Answer booklets or lined paper
- Exam attendance register
- Clock visible to students
- "Silence — Exam in Progress" signs

LESSON 16.2 — MOCK 5 REVIEW: SELF-MARKING AND INDIVIDUAL CONFERENCES

Topic: Final Review, Progress Analysis & Individual Target-Setting

Duration: 40 min | Lesson Type: Review & Conference | Syllabus Ref: 2.1–2.3

Learning Objective:

By the end of this lesson, students will have self-marked their Mock 5, completed their final progress analysis across all mocks, and received an individual conference with their teacher about their final preparation.

Key Vocabulary:

- individual conference (个别谈话 gèbié tánhuà) — one-to-one meeting
- final target (最终目标 zuìzhōng mùbiāo) — the last goal before exams
- achievement (成就 chéngjiù) — something you have done well
- journey (历程 lìchéng) — the path you have travelled
- preparation (准备 zhǔnbèi) — getting ready

Lesson Structure (40 Minutes):

0–5 min | DO NOW + SELF-MARKING

Students receive papers and answer keys. Self-mark (if not teacher-marked) or review teacher marks. Complete Mock 4→5 comparison on progress tracker.

5–15 min | WHOLE-CLASS REVIEW

Teacher presents class data:

- Class average, highest score, range
- Grade distribution
- Comparison with Mock 3 (previous Paper 2)
- Topic analysis: which Paper 2 topics caused most difficulty?

Common errors addressed (quickly):

- Pseudocode: missing ENDFOR/ENDIF
- SQL: forgetting the JOIN condition
- Trace tables: stopping before condition becomes false
- Boolean: NOT gate confusion (0→1, 1→0)

15–35 min | INDIVIDUAL CONFERENCES

While class works on "Final Confidence Checklist" (Worksheet 16A), teacher calls students one by one for 1–2 minute conferences:

- "Your Mock 4 was ____, Mock 5 is ____. That's improvement/drop."

- "Your strongest area is _____. Keep it up."
- "Your priority for final revision is _____."
- "I believe you can achieve _____ if you _____."
- "Any questions? Any worries about the exam?"

Students not in conference complete their checklist independently.

35–40 min | FINAL PLENARY

"You have worked incredibly hard this year. Look at how far you have come."

Teacher projects the class "Progress Journey" chart — visual showing every student's improvement from Mock 1 to now.

"The real exams are just around the corner. You are ready."

Final homework set.

HOMEWORK: Complete the Final Confidence Checklist. Revise your ONE priority topic for 30 minutes. Rest well — you've earned it.

EAL Support:

- Individual conferences adapted for language level
- Teacher uses simple sentences, visual references to tracker
- Checklist available in bilingual format
- Celebrations and encouragement are universal — smiles and thumbs up!

Differentiation:

Emerging:

- Conference is longer (3–4 minutes) with more support
- Simplified checklist with visuals
- Focus on building confidence, not just setting targets
- Teacher provides a printed "revision priority card" with 2–3 specific things to do before the exam

Developing:

- Standard conference (1–2 minutes)
- Full checklist
- Two specific revision priorities identified

Secure:

- Brief conference (1 minute) — mainly confirmation and stretch
- Full checklist + extension reflection
- Student articulates their own plan before teacher confirms

Resources Needed:

- Marked papers or answer keys
- Full mark scheme for projection
- Worksheet 16A — Final Confidence Checklist

- Class progress journey chart (teacher-prepared)
- Individual conference seating (private corner or corridor)
- Revision priority cards (template)

MOCK EXAM 5: IGCSE COMPUTER SCIENCE (0478) PAPER 2

PROBLEM-SOLVING AND PROGRAMMING

Time: 1 hour 45 minutes Maximum Marks: 75

INSTRUCTIONS:

- Answer ALL questions.
- Use black or blue pen. You may use pencil for diagrams and flowcharts.
- For pseudocode, use correct syntax:
 - Assignment: $\leftarrow$ (not =)
 - Keywords: INPUT, OUTPUT, IF, THEN, ELSE, ENDIF, FOR, TO, ENDFOR, WHILE, ENDWHILE, REPEAT, UNTIL
 - One instruction per line
- For trace tables, show ALL working. Complete EVERY row.
- For SQL, use standard SQL syntax.
- Write your answer for Question 7 (the 15-mark scenario) in the spaces provided. Use the planning section before you write your answer.

QUESTION 1 [8 marks]

Algorithms and Pseudocode

(a) Write an algorithm in pseudocode that asks the user to input a password.

The password must be at least 8 characters long. If the password is too short, the program should output "Too short" and ask again. The program should keep asking until a valid password is entered, then output "Password accepted". (4)

(b) Write an algorithm in pseudocode that calculates the sum of all odd numbers between 1 and 100 (inclusive) and outputs the result.

You must use a loop. (4)

QUESTION 2 [8 marks]

Trace Tables

(a) Trace the following algorithm using a trace table: (5)

$x \leftarrow 3$

$y \leftarrow 1$

WHILE $x < 20$

$y \leftarrow y \times x$

$x \leftarrow x + 3$

ENDWHILE

OUTPUT y

Your trace table should include columns for: x, y, x < 20?

(b) Study this pseudocode: (3)

FOR i ← 1 TO 3

FOR j ← 1 TO 3

OUTPUT i × j

ENDFOR

ENDFOR

How many times will the OUTPUT statement be executed?

What will be the first three values output?

What will be the last value output?

QUESTION 3 [8 marks]

Flowcharts

(a) Draw a flowchart for an algorithm that: (5)

- Inputs a temperature reading
- If temperature is 30 or above, outputs "Hot"
- If temperature is between 20 and 29, outputs "Warm"
- If temperature is between 10 and 19, outputs "Cool"
- If temperature is below 10, outputs "Cold"
- The algorithm repeats until the user inputs -999

Use standard flowchart symbols: oval (start/end), rectangle (process), diamond (decision), parallelogram (input/output).

(b) State one advantage and one disadvantage of using a flowchart rather than pseudocode to represent an algorithm. (3)

QUESTION 4 [8 marks]

Programming — Python

(a) What is the output of the following Python code? (2)

```
total = 0
for i in range(1, 6):
    if i % 2 == 0:
        total = total + i
print(total)
```

(b) The following Python function contains three errors. Identify and correct each error. (3)

```
def calculate_average(num1 num2 num3)
```

```
total = num1 + Num2 + num3
```

```
return total / 3
```

```
print("The average is " total)
```

- (c) Write a Python function called `find_max` that takes a list of numbers as a parameter and returns the largest number in the list.

Do not use the built-in `max()` function. (3)

QUESTION 5 [10 marks]

Databases and SQL

A hospital database has the following tables:

Patient(PatientID, FirstName, LastName, DateOfBirth, BloodType)

Appointment(AppointmentID, PatientID, DoctorID, Date, Time, Department)

Doctor(DoctorID, FirstName, LastName, Specialisation, Phone)

- (a) Write an SQL query to display the first name, last name, and blood type of all patients. The results should be ordered by last name alphabetically.

(2)

- (b) Write an SQL query to find all appointments in the 'Cardiology' department on '2024-06-15'. Display the patient's first name and last name, and the appointment time. (3)

- (c) Write an SQL query to count how many appointments each doctor has. Display the doctor's last name and the number of appointments.

(3)

- (d) A new patient is registered. Write an SQL statement to insert the following data into the Patient table:

PatientID: P10045, FirstName: 'Li', LastName: 'Wei',

DateOfBirth: '2008-03-15', BloodType: 'O+' (2)

QUESTION 6 [8 marks]

Boolean Logic

- (a) Draw a logic circuit for the Boolean expression: (3)

$Q = (A \text{ OR } B) \text{ AND } (\text{NOT } C)$

Label all inputs, gates, and the output.

- (b) Complete the truth table for $Q = (A \text{ OR } B) \text{ AND } (\text{NOT } C)$: (3)

A	B	C	NOT C	A OR B	Q
0	0	1			
0	1	0			
0	1	1			
1	0	0			

| 1 | 0 | 1 | | |
| 1 | 1 | 0 | | |

(c) Simplify the following Boolean expression using Boolean algebra.

Show each step and state the law used. (2)

$$X = A \text{ OR } (A \text{ AND } B)$$

QUESTION 7 [15 marks]

Scenario Question — Online Shopping System

SCENARIO

ShopSmart is an online shopping website. Customers can browse products, add them to a shopping cart, and place orders. The system needs to:

- Store product information: ProductID, Name, Category, Price, StockLevel
- Store customer information: CustomerID, Name, Email, Address
- Store order information: OrderID, CustomerID, OrderDate, TotalAmount
- Store order line information: OrderLineID, OrderID, ProductID, Quantity
- When an order is placed, the stock level of each product must be reduced
- If a product's stock level falls below 10, the system should alert the manager to reorder
- Customers can search for products by name or category

PLANNING SPACE (use this before writing your answer)

(a) Design a relational database for this system. [5 marks]

Tables needed:

Primary keys:

Foreign keys:

Relationships:

(b) Write pseudocode for the process of placing an order. [6 marks]

Inputs:

Processes:

Outputs:

Error handling:

(c) Explain two security measures that should be implemented. [2 marks]

Measure 1:

Measure 2:

(d) Describe how you would test the order placement process. [2 marks]

Test 1:

Test 2:

NOW WRITE YOUR ANSWER

- (a) Design a relational database for this system. Show the tables with field names, data types, and primary keys. Explain how the tables are related. (5)
- (b) Write pseudocode for the process of placing an order. Your pseudocode should:
- Input the CustomerID and the products they want to order
 - Check that each product is in stock
 - Calculate the total amount
 - Reduce the stock level for each product ordered
 - Output a confirmation message with the order details
 - If any product is out of stock, output a suitable message
- (6)
- (c) Explain two security measures that should be implemented to protect customer data. Give a reason for each measure. (2)
- (d) Describe how you would test the order placement process. Include at least two different types of test data with reasons. (2)

QUESTION 8 [10 marks]

Program Design and Testing

- (a) Describe what is meant by black-box testing and white-box testing. Give one difference between them. (3)
- (b) A program is being developed to calculate the postage cost for parcels. The cost is based on weight:
- Up to 1 kg: \$5
 - Over 1 kg up to 5 kg: \$10
 - Over 5 kg up to 10 kg: \$15
 - Over 10 kg: \$20 + \$2 for each kg over 10
- (i) Identify three different types of test data you would use to test this program, giving a specific example for each type. (3)
- (ii) For the test case of a 12 kg parcel, what is the expected output? Show your working. (2)
- (iii) Explain why boundary testing is particularly important for this program. (2)

END OF PAPER

MARK SCHEME — MOCK EXAM 5 PAPER 2

QUESTION 1 [8 marks]

- (a) password ← ""

```

WHILE LENGTH(password) < 8
  password ← INPUT "Enter password:"
  IF LENGTH(password) < 8 THEN
    OUTPUT "Too short"
  ENDIF
ENDWHILE

```

```

OUTPUT "Password accepted"

```

(1 mark loop/condition, 1 mark input, 1 mark IF/OUTPUT inside loop,
1 mark final output, 1 mark correct condition check)

Accept REPEAT...UNTIL equivalent. Max 4 marks.

(b) total ← 0

```

FOR num ← 1 TO 100
  IF num MOD 2 = 1 THEN
    total ← total + num
  ENDIF
ENDFOR

```

```

OUTPUT total

```

Alternative: FOR num ← 1 TO 100 STEP 2

(1 mark total init, 1 mark loop, 1 mark odd check, 1 mark correct output)

Answer: 2500

QUESTION 2 [8 marks]

(a) Trace table (5 marks):

x	y	x < 20?
3	1	True
6	3	True
9	18	True
12	162	True
15	1944	True
18	29160	True
21	524880	False

OUTPUT: 524880

(0.5 marks per correct row, max 4 + 1 mark output)

(b) (i) OUTPUT executed 9 times (3×3) (1)

(ii) First three: 1, 2, 3 (1)

(iii) Last value: 9 (1)

QUESTION 3 [8 marks]

(a) Flowchart marking (5 marks):

- Start and End symbols (oval) correctly used (1)
- Input temperature (parallelogram) (1)
- Decision diamond with temperature ranges correctly structured (1)
- Output for each temperature category (1)
- Loop back to input until -999 entered (1)

Accept any logically correct flowchart using correct symbols.

(b) Advantage: visual / easy to understand flow of logic / good for non-programmers / shows branching clearly (1 + development 1)

Disadvantage: takes up more space than pseudocode / harder to represent complex logic / difficult to convert directly to code (1 + dev 1)

Max 3 marks

QUESTION 4 [8 marks]

(a) Output: 6 (0+2+4 = 6, only even numbers 2 and 4 added) (2 marks)

(b) Error 1: Missing commas between parameters → (num1, num2, num3) (1)

Error 2: Num2 should be lowercase → num2 (case sensitive) (1)

Error 3: print statement after return — never executed (1)

OR: print using wrong concatenation syntax

Max 3 marks

(c) def find_max(numbers):

largest = numbers[0]

for num in numbers:

if num > largest:

largest = num

return largest

(1 mark def/name/param, 1 mark initialisation, 1 mark loop+comparison, 1 mark return)

QUESTION 5 [10 marks]

(a) SELECT FirstName, LastName, BloodType FROM Patient

ORDER BY LastName ASC;

(1 mark SELECT+FROM, 1 mark ORDER BY)

(b) SELECT Patient.FirstName, Patient.LastName, Appointment.Time

FROM Patient INNER JOIN Appointment

ON Patient.PatientID = Appointment.PatientID

WHERE Appointment.Department = 'Cardiology'

AND Appointment.Date = '2024-06-15';

(1 mark JOIN, 1 mark WHERE, 1 mark correct columns)

(c) `SELECT Doctor.LastName, COUNT(Appointment.AppointmentID) AS NumAppointments
FROM Doctor INNER JOIN Appointment
ON Doctor.DoctorID = Appointment.DoctorID
GROUP BY Doctor.DoctorID, Doctor.LastName;`

(1 mark COUNT, 1 mark GROUP BY, 1 mark JOIN)

(d) `INSERT INTO Patient (PatientID, FirstName, LastName, DateOfBirth, BloodType)
VALUES ('P10045', 'Li', 'Wei', '2008-03-15', 'O+');`

(1 mark INSERT INTO with table/columns, 1 mark VALUES)

QUESTION 6 [8 marks]

(a) Logic circuit (3 marks):

- A and B into OR gate (1)
- C into NOT gate (1)
- Output of OR and NOT into AND gate → Q (1)

(b) Truth table (3 marks):

A	B	C	NOT C	A OR B	Q
0	0	0	1	0	0
0	0	1	0	0	0
0	1	0	1	1	1
0	1	1	0	1	0
1	0	0	1	1	1
1	0	1	0	1	0
1	1	0	1	1	1
1	1	1	0	1	0

(0.5 marks per 2 correct rows, max 3)

(c) $X = A \text{ OR } (A \text{ AND } B)$

$X = (A \text{ OR } A) \text{ AND } (A \text{ OR } B) \leftarrow \text{distributive law (1)}$

$X = A \text{ AND } (A \text{ OR } B) \leftarrow \text{idempotent law (A OR A = A)}$

$X = A \leftarrow \text{absorption law (1)}$

Accept: direct application of absorption law for full marks.

QUESTION 7 [15 marks]

(a) Database design (5 marks):

Product: ProductID (PK, STRING/INTEGER), Name (STRING), Category (STRING),
Price (DECIMAL), StockLevel (INTEGER) (1)

Customer: CustomerID (PK, STRING/INTEGER), Name (STRING), Email (STRING),
Address (STRING) (1)

Order: OrderID (PK), CustomerID (FK), OrderDate (DATE),

TotalAmount (DECIMAL) (1)

OrderLine: OrderLineID (PK), OrderID (FK), ProductID (FK),

Quantity (INTEGER) (1)

— Relationships: Product to OrderLine (one-to-many)

Customer to Order (one-to-many)

Order to OrderLine (one-to-many) (1)

— All related by foreign keys (1)

Max 5 marks

(b) Pseudocode for placing order (6 marks):

customerID ← INPUT

orderTotal ← 0

moreItems ← TRUE

WHILE moreItems = TRUE

productID ← INPUT "Enter product ID:"

quantity ← INPUT "Enter quantity:"

// Find product

FOR i ← 0 TO Products.length - 1

IF Products[i].ProductID = productID THEN

IF Products[i].StockLevel >= quantity THEN

Products[i].StockLevel ← Products[i].StockLevel - quantity

orderTotal ← orderTotal + (Products[i].Price × quantity)

OUTPUT "Added to order"

ELSE

OUTPUT "Not enough stock"

ENDIF

ENDIF

ENDFOR

response ← INPUT "Add more items? (Y/N)"

IF response = "N" THEN

moreItems ← FALSE

ENDIF

ENDWHILE

// Save order

OUTPUT "Order total:", orderTotal

OUTPUT "Order confirmed"

(1 mark input customer, 1 mark loop for items, 1 mark stock check,
1 mark reduce stock, 1 mark calculate total, 1 mark output,
1 mark handle out of stock)

Max 6 marks. Accept equivalent logic.

(c) Security measures (2 marks):

Any 2: encryption of customer data/passwords, SSL/HTTPS for data
transmission, access controls/user authentication, input validation
to prevent SQL injection, regular backups, firewall.

(1 mark per measure + reason)

(d) Testing (2 marks):

Normal: valid customer, in-stock products → successful order (1)

Boundary: order exactly 10 items (stock=10) → should work, alert
triggered if stock goes below 10 (1)

Erroneous: invalid product ID → error message (1)

Erroneous: out-of-stock product → suitable message (1)

Max 2 marks (need 2 types with data + reason)

QUESTION 8 [10 marks]

(a) Black-box: tests inputs/outputs without looking at code structure,
tests functionality from user perspective (1)

White-box: examines internal code structure, tests all paths/branches
through code (1)

Difference: black-box treats program as opaque / doesn't see code;
white-box requires knowledge of code structure (1)

Max 3 marks

(b) (i) Three types of test data (3 marks):

— Normal: 3 kg → \$10 (1 + example)

— Boundary: 1 kg → \$5, 5 kg → \$10, 10 kg → \$15 (1 + example)

— Erroneous: -2 kg → error/invalid (1 + example)

(1 mark per type with valid example)

(ii) 12 kg: $\$20 + (2 \times \$2) = \$24$ (1 mark working, 1 mark answer)

(iii) Boundary testing important because costs change at specific
weight thresholds (1). A program might use $\geq$ instead of $>$, causing
wrong cost at exact boundary weights (1). Must verify correct
behaviour at 1, 5, and 10 kg exactly (1). Max 2 marks.

GRADE BOUNDARIES — MOCK EXAM 5 PAPER 2

A*: 69–75 (92–100%)

A: 61–68 (81–91%)
 B: 53–60 (71–80%)
 C: 45–52 (60–70%)
 D: 37–44 (49–59%)
 E: 29–36 (39–48%)
 F: 21–28 (28–38%)
 G: 13–20 (17–27%)
 U: 0–12 (0–16%)

--- WORKSHEET 16A ---

WORKSHEET 16A: FINAL CONFIDENCE CHECKLIST

Name: _____ Date: _____ Final Target Grade: _____

PART A: MY MOCK JOURNEY (complete for both papers)

PAPER 1:

Mock	Mark /75	Grade	Grade boundaries used: A*=69, A=61, B=53, C=45
M1			
M2			
M4			

PAPER 2:

Mock	Mark /75	Grade	Grade boundaries used: A*=69, A=61, B=53, C=45
M1			
M3			
M5			

PART B: TOPIC CONFIDENCE CHECKLIST

Rate each topic: 3 = Confident 2 = Getting there 1 = Need more practice

Paper 1 Topics:

Topic	Rating	One thing I still need to check
Binary/hex conversions		
File size calculations		
Compression		
CPU architecture		
Memory types		
Storage comparison		
OS functions		
Networks		

Security threats		
Emerging tech		

Paper 2 Topics:

Topic	Rating	One thing I still need to check
Pseudocode writing		
Trace tables		
Flowcharts		
Python programming		
SQL queries		
Database design		
Boolean logic		
Program testing		
15-mark scenario		

PART C: MY EXAM STRATEGY

51. The first thing I will do when I open the exam paper: _____
52. If I get stuck on a question, I will: _____
53. I will check my answers by: _____
54. The topic I will revise tonight: _____

PART D: PERSONAL MESSAGE

Write a message to yourself to read before the real exam:

Dear me,

You have worked hard and you are ready. Remember:

You can do this!

Signed: _____ Date: _____

--- ANSWER KEY: WORKSHEET 16A ---

ANSWER KEY: WORKSHEET 16A — FINAL CONFIDENCE CHECKLIST

This is a reflective worksheet — no "right" answers. Mark for completion and quality of reflection.

MARKING CRITERIA (10 marks):

- Part A completed accurately (3 marks)
 - All mocks filled in (1)
 - Grades correct from boundaries (1)
 - Both Paper 1 and Paper 2 completed (1)
- Part B completed honestly and specifically (3 marks)
 - All topics rated (1)
 - "One thing to check" column has specific notes, not just "revise" (1)
 - Ratings match mock performance (approximately) (1)

- Part C has specific, practical strategies (2 marks)
 - Each strategy is actionable (1)
 - Strategies are appropriate for that student (1)
- Part D is completed with encouraging, personal message (1 mark)
- Overall completion and effort (1 mark)

TEACHER NOTES:

- Collect before students leave — this is their final planning document.
- Flag any student who has rated everything "1" — they need a confidence-building conversation.
- Flag any student who has rated everything "3" — they may be overconfident and need a gentle reality check.
- Use Part D messages to create a "Good Luck" display for the class.

UNIT 6 OVERVIEW

Final Preparation & Exam Readiness

Unit 6 spans Weeks 17–18 and represents the final phase before the IGCSE

examinations. These two weeks are deliberately light on new content and heavy on confidence-building, exam technique, and emotional preparation.

The goal is NOT to cram new information but to:

55. Consolidate existing knowledge through student-led revision

2. Build exam confidence and reduce anxiety

3. Ensure every student has clear exam day strategies

4. Finish the course on a positive, motivating note

Week 17: Last-Minute Revision Clinic

Week 18: Final Sessions & Exam Preparation

WEEK 17: LAST-MINUTE REVISION CLINIC

OVERVIEW:

Week 17 is student-driven. Students vote on which topics they want to

cover, and the teacher delivers rapid revision sessions on the most popular choices. The week also covers essential exam day strategies and includes confidence-building activities.

LESSON 17.1 — STUDENT-CHOSE REVISION & EXAM DAY TIPS

Topic: Student-Led Revision & Exam Strategy

Duration: 40 min | Lesson Type: Student-Led Clinic | Syllabus Ref: All

Learning Objective:

By the end of this lesson, students will have received targeted revision on their most-requested topics and learned practical strategies for managing exam day challenges.

Key Vocabulary:

- vote (投票 *tóupiào*) — to choose by marking a ballot
- rapid revision (快速复习 *kuàisù fùxí*) — quick review of key points
- exam strategy (考试策略 *kǎoshì cèlüè*) — plan for doing well in exam
- time management (时间管理 *shíjiān guǎnlǐ*) — using time well
- if stuck (如果卡住 *rúguǒ kǎ zhù*) — what to do when you don't know

Lesson Structure (40 Minutes):

0–3 min | DO NOW — VOTE

Students receive 3 sticky dots. They place dots on the "Topic Voting Board" (displayed on whiteboard) next to their most-wanted revision topics.

Topic options (with Chinese translations):

- Binary/Hex conversions (二进制/十六进制)
- File size calculations (文件大小计算)
- CPU and F-E cycle (CPU 与取指周期)
- Networks and security (网络与安全)
- Pseudocode writing (伪代码编写)
- Trace tables (追踪表)
- SQL queries (SQL 查询)
- Boolean logic (布尔逻辑)
- 15-mark scenario (15 分情景题)

Top 3 topics identified by dot count.

3–12 min | RAPID REVISION — TOPIC 1 (most votes)

Teacher delivers a focused 9-minute revision:

- 2 min: Key facts (3–4 bullet points on board)

- 3 min: Worked example (teacher models on board)
- 2 min: Students try one quick question (mini-whiteboards)
- 2 min: Check and correct

High energy, fast pace, no time for distractions.

12-22 min | RAPID REVISION — TOPIC 2

Same structure — different topic.

22-32 min | RAPID REVISION — TOPIC 3

Same structure — different topic.

32-40 min | EXAM DAY TIPS

Teacher shares practical exam strategies:

"WHAT TO DO WHEN YOU OPEN THE PAPER:"

- Read the instructions carefully (different papers have different rules)
- Skim all questions first — note the mark allocation
- Start with a question you feel confident about (builds momentum)
- Show ALL working — method marks save you when answers are wrong

"WHAT TO DO IF YOU'RE STUCK:"

- Don't panic! Move on and come back later
- Write down what you DO know — even partial answers get marks
- For calculation questions: write the formula — that's a mark!
- For "explain" questions: use "because" — explain WHY, not just WHAT
- For pseudocode: write the structure first (IF...ENDIF, loop...ENDLOOP), then fill in the details

"TIME MANAGEMENT:"

- 75 marks in 105 minutes = 1.4 minutes per mark
- 6-mark question = about 8 minutes
- If a 1-mark question is taking 5 minutes, MOVE ON
- Leave 10 minutes at the end to check
- Check: Have I answered EVERY question? Even a guess is better than blank

Students write their top 3 exam tips on a card to keep in their pocket on exam day.

EAL Support:

- Voting board uses icons as well as text
- Exam tips written in English and Chinese on display
- Teacher models exam tips with visual gestures (pointing to clock, turning page, deep breath)
- Students can write pocket card in Chinese if preferred

Differentiation:

Emerging:

- During rapid revision, LSA provides extra support at their desk
- Given pre-written exam tips card (just personalise)
- Encouraged to focus on recall questions (definitions, naming)

Developing:

- Full participation in rapid revision
- Write own exam tips card
- Ask clarifying questions about exam strategy

Secure:

- Full participation
- During exam tips section, share own strategies with class
- Help compose exam tips for peers

Resources Needed:

- Topic Voting Board on whiteboard (pre-drawn)
- Coloured sticky dots (3 per student)
- Mini-whiteboards and pens
- Prepared revision content for all 9 topics (teacher prepares in advance)
- "Exam Tips" display poster (bilingual)
- Pocket cards for students to write tips on

LESSON 17.2 — Q&A SESSION & CONFIDENCE BUILDING

Topic: Student Questions, Quick-Fire Practice & Confidence Activity

Duration: 40 min | Lesson Type: Q&A Workshop | Syllabus Ref: All

Learning Objective:

By the end of this lesson, students will have had their revision questions answered, practised 10 quick-fire exam questions, and participated in a confidence-building activity.

Key Vocabulary:

- Q&A (问答 wèndá) — questions and answers
- quick-fire (快速问答 kuàisù wèndá) — fast questions
- confidence (信心 xìnxīn) — belief that you can do it
- statement (陈述 chénnshù) — a sentence saying something
- I know I can... (我知道我能... wǒ zhīdào wǒ néng...) — positive phrase

Lesson Structure (40 Minutes):

0–10 min | STUDENT QUESTIONS

"Open floor" — students ask any question from their independent revision.

Teacher answers or invites class to answer.

Common themes addressed:

- "How much working do I need to show?" → ALL of it. Every step.
- "What if I forget pseudocode syntax?" → Write what makes sense — examiners award marks for correct logic even if syntax is imperfect.
- "Do I need to memorise all Boolean laws?" → Know: $A+0=A$, $A\cdot 1=A$, $A+\bar{A}=1$, $A\cdot\bar{A}=0$, distributive, De Morgan's. The rest can be derived.
- "How do I approach the 15-mark question?" → PLAN FIRST. 5 minutes planning saves 10 minutes rewriting.

10–25 min | QUICK-FIRE 10 (5 Paper 1, 5 Paper 2)

Students number 1–10 in their books. Teacher reads questions aloud, students write answers. 30 seconds per question. Fast pace!

Paper 1 Questions:

56. Convert 1110 binary to denary. (1)
57. What is the difference between RAM and ROM? (1)
58. Name two network topologies. (1)
59. What does a firewall do? (1)
60. Name one emerging technology. (1)

Paper 2 Questions:

61. What is the output? $x \leftarrow 5$, $y \leftarrow x \times 2$, OUTPUT y (1)
62. Write the first line of an SQL query to select names from a Customers table where City = 'Shenzhen'. (1)
63. What does this Python code output? `print(10 % 3)` (1)
64. Draw the symbol for an AND gate. (1)
65. In a trace table, what should you write in the "condition" column? (1)

Answers revealed — students self-mark. Celebrate 8+/10!

25–35 min | CONFIDENCE-BUILDING: "I KNOW I CAN..."

Each student writes 3 "I know I can..." statements:

- "I know I can convert binary to hex because I've practised it 20 times."
- "I know I can write pseudocode because I know the structure."
- "I know I can do well in this exam because I have worked hard."

Students share with a partner, then 3 volunteers share with the class.

Teacher adds: "I know YOU can all do this because I have seen your progress this year."

35–40 min | FINAL CHECKLIST DISTRIBUTION

Distribute Worksheet 17A: Final Exam Checklist.

HOMEWORK: Use the checklist tonight. Tick off each item as you do it.

Bring to next lesson (the final lesson).

EAL Support:

- Q&A: teacher repeats question in simpler English before answering
- Quick-fire: questions shown on board as well as spoken
- Confidence activity: students can write statements in Chinese first then translate with teacher/peer help
- "I know I can..." frame provided: "I know I can [topic] because [reason]."

Differentiation:

Emerging:

- During Q&A, teacher or LSA checks they understand answers
- Quick-fire: can write answers in Chinese if English term forgotten
- Confidence statements: teacher helps compose 1 strong statement

Developing:

- Full participation
- Write and share 3 statements independently

Secure:

- Full participation
- Help answer peers' questions during Q&A
- Model confident statements for class

Resources Needed:

- Whiteboard for Q&A
- 10 quick-fire questions (printed and projected)
- Answer key for quick-fire
- "I know I can..." template on board
- Worksheet 17A — Final Exam Checklist

--- WORKSHEET 17A ---

WORKSHEET 17A: FINAL EXAM CHECKLIST & EXAM DAY PREPARATION GUIDE

Name: _____ Date: _____ Exam Date: _____

TONIGHT'S REVISION CHECKLIST

Tick each item as you complete it:

- [] 1. Review my Personal Performance Tracker — focus on my 3 weakest topics
- [] 2. Complete 5 quick questions on topic 1: _____
- [] 3. Complete 5 quick questions on topic 2: _____
- [] 4. Check I know all pseudocode keywords (list them below):
- [] 5. Check I know the SQL keywords: SELECT, FROM, WHERE, JOIN, ORDER BY, GROUP BY, INSERT, UPDATE, DELETE
- [] 6. Review the Boolean algebra laws (write the 4 most important):
- [] 7. Look at one 15-mark scenario from a past paper — read the mark scheme
- [] 8. Pack my exam bag (see below)

☐ 9. Go to bed early — aim for 8 hours sleep

☐ 10. Set TWO alarms!

EXAM DAY CHECKLIST — WHAT TO BRING

ESSENTIAL (you cannot take the exam without these):

☐ Black or blue pens (bring 2-3 — they can run out!)

☐ Pencil (for diagrams and flowcharts)

☐ Ruler

☐ Eraser

☐ Calculator (for Paper 1 file size calculations)

☐ Student ID card

☐ Exam timetable/confirmation

RECOMMENDED:

☐ Watch (analogue — no smart features)

☐ Tissues

☐ Water bottle (clear, no label)

☐ Small snack for between papers

DO NOT BRING:

☐ Mobile phone (must be switched off and left outside)

☐ Smart watch

☐ Notes or revision materials

☐ Food that makes a noise or smell

EXAM DAY TIMELINE

The night before:

— Light revision only (30 minutes max) — don't cram!

— Review your "I know I can..." statements

— Pack your bag

— Early bedtime

Morning of exam:

— Wake up with plenty of time

— Eat breakfast (brain needs fuel!)

— Arrive 30 minutes early

— Use the toilet before entering exam room

— Take deep breaths — you ARE ready

During exam:

— Read instructions carefully

— Show ALL working

- Manage your time — 1.4 minutes per mark
- If stuck, move on and come back
- Check every answer before time is up

After exam:

- Don't discuss answers with friends (it causes worry!)
- Focus on the NEXT exam, not the one just done
- Reward yourself with a treat!

EMERGENCY "WHAT IF" PLAN

"What if I forget a formula?"

- Write down what you DO remember. Part marks available.
- For file sizes: think "width × height × depth" for images

"What if I don't understand a question?"

- Read it THREE times slowly
- Underline key words
- Try to rephrase it in your own words
- If still stuck, mark it and move on — come back later

"What if I run out of time?"

- Make sure every question has ANSWER — even a guess is better than blank
- For calculation questions: write the formula — that's worth marks!
- For "explain" questions: write anything relevant — partial marks!

"What if I panic?"

- STOP. Put down your pen.
- Take 3 deep breaths (in through nose, out through mouth)
- Drink some water
- Remind yourself: "I have prepared. I can do this."
- Start with an easy question to build confidence

MY PERSONAL "I KNOW I CAN" STATEMENTS

66. I know I can _____
because _____
67. I know I can _____
because _____
68. I know I can _____
because _____

--- ANSWER KEY: WORKSHEET 17A ---

ANSWER KEY: WORKSHEET 17A — FINAL EXAM CHECKLIST

This is a planning/reflection worksheet — no "correct" answers.

Mark for completion and quality.

MARKING CRITERIA (10 marks):

- Revision checklist: at least 6 items ticked/attempted (3 marks)
- Exam bag checklist: essentials identified (2 marks)
- "What if" plan read and understood (2 marks — teacher initials)
- Personal statements completed meaningfully (2 marks)
- Overall presentation and effort (1 mark)

TEACHER NOTES:

- This worksheet is designed to be PRACTICAL — students should literally use it the night before the exam.
- Photocopy the "What to Bring" section for students to keep in their bag.
- The "What If" plan is particularly important for anxious students.
- Collect at the start of Lesson 18.1 — review and return before they leave.

ANSWER KEY: QUICK-FIRE 10 (from Lesson 17.2)

69. 14 (8+4+2)

70. RAM is volatile/read-write; ROM is non-volatile/read-only (accept any valid difference)

3. Any two: star, bus, ring, mesh, tree

71. Monitors network traffic / blocks unauthorised access / filters packets

72. Any valid: AI, IoT, blockchain, cloud computing, VR/AR, quantum computing, 5G, autonomous vehicles

73. 10

7. SELECT Name FROM Customers WHERE City = 'Shenzhen';

(accept without semicolon, accept without capitalisation)

74. 1 (10 ÷ 3 = 3 remainder 1)

75. D-shaped symbol with two inputs on left, one output on right

76. Whether the loop condition is true or false / whether the loop continues

WEEK 18: FINAL SESSIONS & EXAM PREPARATION

OVERVIEW:

Week 18 is the final week of the entire Year 10 Computer Science course.

These two lessons are about finishing strong: giving students the final

tips, strategies, and confidence they need to walk into the exam room feeling prepared and positive. The week ends with a motivational close that recognises every student's progress and effort.

LESSON 18.1 — FINAL TIPS AND STRATEGIES

Topic: Paper-Specific Exam Strategies & Last-Minute Revision

Duration: 40 min | Lesson Type: Strategy Workshop | Syllabus Ref: All

Learning Objective:

By the end of this lesson, students will know the specific strategies for each paper type and have identified their final revision priorities.

Key Vocabulary:

- strategy (策略 cèlüè) — a plan for success
- technique (技巧 jìqiǎo) — a way of doing something well
- terminology (术语 shùyǔ) — special words used in a subject
- working (步骤 bùzhòu) — showing how you got the answer
- priority (优先 yōuxiān) — most important thing to do

Lesson Structure (40 Minutes):

0–3 min | DO NOW

Students receive their marked Worksheet 17A (Final Checklist) back.

On a slip of paper, they write:

- One thing they learned from doing the checklist
- One question they still have

Teacher collects while setting up.

3–15 min | PAPER 1 STRATEGIES

Teacher presents Paper 1 specific advice:

"PAPER 1 — THEORY: HOW TO MAXIMISE YOUR MARKS"

77. SHOW ALL WORKING (显示所有计算过程)

- For binary conversions: write the place values (128, 64, 32...)
- For file sizes: write the formula BEFORE substituting numbers
- Even if your final answer is wrong, method marks can save you!

78. USE CORRECT TERMINOLOGY (使用正确术语)

- "Volatile" not "disappears when you turn it off"
- "Lossy compression" not "makes files smaller"
- "Router" not "the thing that connects to the internet"
- Examiners reward precise technical language

79. DRAW CLEAR DIAGRAMS (画出清晰图表)

- Use a ruler for network topologies
- Label every component in CPU diagrams

- Arrows should be clear and point in the right direction

80. EXPLAIN, DON'T JUST NAME (解释, 不要只命名)

- "Lossy compression removes data permanently" → 1 mark
- "Lossy compression removes data permanently, which means the original file cannot be perfectly reconstructed. This is acceptable for photographs where some detail loss is not noticeable." → 3 marks
- The word "because" is your friend!

81. ANSWER EVERY QUESTION (回答每个问题)

- Never leave a blank. Even a guess or partial answer is better than nothing.
- For "state" or "name" questions: list anything relevant.

Students highlight the strategy they find most useful on their printed strategy sheet.

15–28 min | PAPER 2 STRATEGIES

Teacher presents Paper 2 specific advice:

"PAPER 2 — PROBLEM SOLVING: HOW TO MAXIMISE YOUR MARKS"

82. WRITE PSEUDOCODE FIRST, THEN REFINE (先写伪代码, 再完善)

- Start with the structure: FOR...ENDFOR, IF...ENDIF
- Then fill in the details
- Check: have I closed every block with ENDWHILE/ENDIF/ENDFOR?

83. TEST YOUR LOGIC (测试你的逻辑)

- For pseudocode: trace through with simple numbers (0, 1, 2)
- Does it do what it's supposed to do?
- Have you handled edge cases? (empty input, zero, maximum)

84. SHOW ALL WORKING IN TRACE TABLES (追踪表显示所有步骤)

- Every variable, every iteration, every condition check
- Don't stop when you think you know the answer — complete ALL rows
- One mark per correct row is common!

85. SQL: START SIMPLE, THEN ADD (SQL: 从简单开始, 然后添加)

- Start with: SELECT * FROM Table
- Then add WHERE conditions
- Then add JOIN if needed
- Then add ORDER BY or GROUP BY
- Check: have I specified the linking column for JOIN?

86. THE 15-MARK QUESTION: PLAN FIRST! (15 分题 : 先计划 !)

- Spend 5 minutes planning before you write
- Read the scenario TWICE
- Identify: what tables? what pseudocode structure? what security?
- Write bullet points first, then expand
- Check: have I answered ALL parts (a), (b), (c), (d)?

Students highlight their most useful Paper 2 strategy.

28–35 min | LAST-MINUTE REVISION PRIORITIES

Teacher displays the "High-Value Topic" list:

- Topics that appear every year AND carry lots of marks:

Paper 1: File sizes (6 marks), CPU/F-E cycle (5+ marks),
Networks (4+ marks), Security (4+ marks)

Paper 2: Pseudocode writing (4–6 marks), Trace tables (4–5 marks),
SQL (4+ marks), 15-mark scenario (15 marks!)

"If you only have time to revise 3 things, make them the ones that
always come up and always carry the most marks."

Students write their top 3 priorities on a card.

35–40 min | PLENARY + HOMEWORK

"One word to describe how I'm feeling about the exam" — students share.

Teacher responds supportively to any anxiety.

Distribute Worksheet 18A: Exam Day Checklist & Personal Achievement
Reflection.

HOMEWORK: Complete the reflection sheet. Do your final 30 minutes of
revision on your #1 priority topic. Then STOP and REST.

EAL Support:

- Strategy sheets printed with Chinese translations for key terms
- Teacher models each strategy with a worked example
- Visual demonstrations: showing good vs bad diagram, good vs bad explanation
- "Highlight your most useful strategy" reduces language pressure

Differentiation:

Emerging:

- Given printed strategy sheets to keep (not just projected)
- During Paper 2 section, provided with pseudocode template card
- Priority topics selected by teacher, not student

Developing:

- Full strategy session
- Self-select priorities with teacher guidance

Secure:

- Full session
- Asked to contribute additional tips from their own experience
- Mentor role: help classmates identify their priorities

Resources Needed:

- Printed strategy sheets (Paper 1 and Paper 2) for each student
- Highlighters
- High-Value Topic display
- Priority cards
- Worksheet 18A

LESSON 18.2 — FINAL CONFIDENCE SESSION & FAREWELL

Topic: Personal Target Review, Exam Logistics & Motivational Close

Duration: 40 min | Lesson Type: Final Session | Syllabus Ref: All

Learning Objective:

By the end of this lesson, students will have reviewed their personal targets from the start of the year, recognised their progress, received exam day logistics, and ended the course feeling motivated and confident.

Key Vocabulary:

- target (目标 mùbiāo) — a goal you set
- progress (进步 jìnbù) — how far you have come
- recognise (认可 rènkě) — to see and appreciate
- logistics (后勤 hòuqín) — practical arrangements
- farewell (告别 gàobié) — saying goodbye

Lesson Structure (40 Minutes):

0–3 min | DO NOW — TARGET REVIEW

At the start of the year, students wrote personal targets on a card.

These cards are now returned to them.

"Open your target card. Read what you wrote at the start of Year 10."

Students silently read their original targets.

3–15 min | PROGRESS RECOGNITION

Teacher leads a structured reflection:

"At the start of this year, you set targets for yourself. Let's see how far you've come."

- Target 1: _____ → Did you achieve it? Are you closer?
- Target 2: _____ → What progress have you made?
- Target 3: _____ → What are you most proud of?

Students complete the "Personal Achievement Reflection" section of

Worksheet 18A.

Teacher shares (anonymously) some growth stories:

- "One student started the year finding binary confusing. In Mock 4, they got full marks on binary conversions. That's incredible growth."
- "Another student couldn't write a single line of pseudocode in September. In Mock 5, they wrote a complete 6-mark algorithm. That's what hard work looks like."

Students invited to share their own achievements (voluntary).

15–25 min | EXAM DAY LOGISTICS

Teacher covers practical arrangements:

- Exam dates, times, and locations (displayed on board)
- Arrival time (30 minutes before)
- What to bring (pen, pencil, ruler, calculator, ID)
- What NOT to bring (phone, smart watch, notes)
- What to do if late or ill (contact school immediately)
- Which room / seat number
- Who to ask if you have a problem

Students write key details on their checklist.

25–35 min | MOTIVATIONAL CLOSE: "YOU ARE READY"

Teacher delivers final motivational message:

"This time last year, many of you had never written a line of pseudocode. You didn't know what a trace table was. You couldn't convert binary to hex. You'd never written an SQL query. Now? You can do ALL of those things. I've seen it. Your mocks prove it.

The exam is not a mystery. It's not a trap. It's an opportunity to show everything you've learned. And you have learned SO much. When you walk into that exam room, hold your head high. You are prepared. You are capable. You are ready.

You've got this."

Final confidence activity: Whole class stands. Together:

"I AM PREPARED." (class repeats)

"I AM CAPABLE." (class repeats)

"I AM READY." (class repeats)

"I'VE GOT THIS!" (class shouts!)

Optional: class photo, group cheer, or tradition (e.g., everyone writes

their name on a "Class of 2024" poster).

35–40 min | FAREWELL AND GOOD LUCK

Teacher distributes:

- Final exam checklist (last-minute reminder)
- Small token if available (e.g., "Good Luck" card, revision sweet)
- Contact details: "Email me if you have questions before the exam"

Final individual goodbyes as students leave.

"Good luck. You've earned it."

EAL Support:

- Exam logistics displayed visually as well as verbally
- Key information written on board in English and Chinese
- Motivational message uses simple, powerful language
- Physical gestures (standing together, clapping, cheering) transcend language

Differentiation:

Emerging:

- Original targets reviewed sensitively — focus on growth, not gaps
- Given extra encouragement during farewell
- Teacher ensures they know exactly where to go and what to bring

Developing:

- Standard reflection and logistics
- Encouraged to share achievements

Secure:

- Standard reflection
- Asked to help motivate classmates during the close
- May be given a "buddy" to support before the exam

Resources Needed:

- Original target cards (kept from start of year)
- Worksheet 18A — Exam Day Checklist & Personal Achievement Reflection
- Exam logistics display (dates, times, rooms)
- "Class of 2024" poster or memento (optional)
- Tokens/cards (optional)

--- WORKSHEET 18A ---

WORKSHEET 18A: EXAM DAY CHECKLIST & PERSONAL ACHIEVEMENT REFLECTION

Name: _____ Date: _____

PART A: MY TARGET REVIEW

My targets from the start of Year 10:

87. _____

88. _____

89. _____

Have I achieved these targets?

Target 1: ☐ Yes ☐ Partly ☐ Not yet Evidence: _____

Target 2: ☐ Yes ☐ Partly ☐ Not yet Evidence: _____

Target 3: ☐ Yes ☐ Partly ☐ Not yet Evidence: _____

PART B: PERSONAL ACHIEVEMENT REFLECTION

90. The topic I knew NOTHING about at the start of the year but now understand is:

2. The mock result I am most proud of is:

Because: _____

3. The skill I have developed most this year is:

4. The thing I found hardest this year was:

How I overcame it (or am overcoming it): _____

91. One piece of advice I would give to next year's Year 10 students:

6. My personal achievement statement:

"This year, I have grown from _____ to _____
_____. I am proud of myself because _____."

PART C: EXAM DAY LOGISTICS

My Paper 1 exam: Date: _____ Time: _____ Room: _____

My Paper 2 exam: Date: _____ Time: _____ Room: _____

I need to arrive by: _____

I will travel to school by: _____

If I have a problem on exam day, I will contact: _____

PART D: FINAL "I KNOW I CAN" STATEMENTS

Write three statements. Read these before you go into the exam room.

92. I know I can _____
because _____

93. I know I can _____
because _____

94. I know I can _____
because _____

FINAL MESSAGE TO MYSELF

Dear me,

You have worked so hard this year. Whatever happens in the exam, remember:

I believe in you.

Signed: _____ Date: _____

--- ANSWER KEY: WORKSHEET 18A ---

ANSWER KEY: WORKSHEET 18A — EXAM DAY CHECKLIST & ACHIEVEMENT REFLECTION

This is a reflective worksheet — no "correct" answers. Mark for completion, quality of reflection, and evidence of self-awareness.

MARKING CRITERIA (10 marks):

- Part A completed with honest self-assessment (2 marks)
- Part B: all 6 sections completed with specific, meaningful content (4 marks — 1 per 2 sections, max 4)
- Part C: exam logistics filled in correctly (2 marks)
- Part D: three positive, specific statements (1 mark)
- Final message is encouraging and personal (1 mark)

TEACHER NOTES:

- This worksheet is a keepsake — encourage students to hold onto it.
- Part B, Question 5 (advice for next year) can be shared with the next Year 10 cohort as authentic student voice.
- Collect and photocopy for your records — wonderful evidence of student growth for reports/parent meetings.
- Return original to student as a memento of their Year 10 journey.

END OF CURRICULUM DOCUMENT

Victoria Park Academy Shenzhen — Year 10

IGCSE Computer Science (0478)

Semester 2, Weeks 10–18

SUMMARY OF CONTENTS:

Week 10: Mock 3 Review & Gap Analysis Preparation

- Lesson 10.1: Mock 3 Paper 2 Error Analysis
- Lesson 10.2: Compiling the Gap Analysis
- Worksheet 10A: Error Analysis Sheet + Answer Key
- Worksheet 10B: Personal Performance Tracker + Answer Key

Week 11: Individual Gap Analysis

- Lesson 11.1: Data-Driven Revision & Grouping
- Lesson 11.2: Targeted Revision — Three Groups Running
- Worksheet 11A: Group Revision Packs (Sets A, B, C) + Answer Key
- Worksheet 11B: Personal Revision Plan + Answer Key

Week 12: Targeted Revision — Paper 1 Weak Areas

- Lesson 12.1: Whole-Class Paper 1 Revision
- Lesson 12.2: Targeted Paper 1 Practice
- Worksheet 12A: Differentiated Paper 1 Revision (Sets A, B, C) + Key
- Worksheet 12B: Quick-Fire Practice + Answer Key

Week 13: Targeted Revision — Paper 2 Weak Areas

- Lesson 13.1: Whole-Class Paper 2 Revision
- Lesson 13.2: Targeted Paper 2 Practice
- Worksheet 13A: Differentiated Paper 2 Revision (Sets A, B, C) + Key
- Worksheet 13B: 15-Mark Scenario Practice + Answer Key

Week 14: Mixed Practice & Consolidation

- Lesson 14.1: Timed Paper 1 Practice (30 min)
- Lesson 14.2: Timed Paper 2 Practice (30 min)
- Worksheet 14A: Mixed Paper 1 Practice Paper + Answer Key
- Worksheet 14B: Mixed Paper 2 Practice Paper + Answer Key

Week 15: Mock Exam 4 — Paper 1

- Lesson 15.1: Full Paper 1 Mock Examination (75 marks, 1h 45min)
- Lesson 15.2: Mock 4 Review & Progress Tracking
- MOCK EXAM 4: Complete Paper 1 (16 questions, 75 marks) + Full Mark Scheme
- Grade Boundaries included
- Worksheet 15A: Mock Progress Tracker (Paper 1 Journey) + Answer Key

Week 16: Mock Exam 5 — Paper 2

- Lesson 16.1: Full Paper 2 Mock Examination (75 marks, 1h 45min)
- Lesson 16.2: Mock 5 Review & Individual Conferences
- MOCK EXAM 5: Complete Paper 2 (8 questions, 75 marks) + Full Mark Scheme

- Grade Boundaries included
- Worksheet 16A: Final Confidence Checklist + Answer Key

Week 17: Last-Minute Revision Clinic

- Lesson 17.1: Student-Chosen Revision & Exam Day Tips
- Lesson 17.2: Q&A Session & Confidence Building
- Worksheet 17A: Final Exam Checklist & Exam Day Guide + Answer Key
- Quick-Fire 10 Questions + Answers

Week 18: Final Sessions & Exam Preparation

- Lesson 18.1: Final Tips & Strategies (Paper 1 & Paper 2)
- Lesson 18.2: Final Confidence Session & Farewell
- Worksheet 18A: Exam Day Checklist & Personal Achievement Reflection + Key

TOTAL DELIVERABLES:

- 18 complete lesson plans (40 minutes each)
- 18 student worksheets with answer keys
- 2 full mock examination papers (75 marks each)
- 2 complete mark schemes with grade boundaries
- 4 differentiated worksheet sets (3 versions each)
- Progress trackers and confidence-building materials
- Exam day preparation guides

All materials include:

- EAL support with bilingual keywords (EN/CN/pinyin)
- Differentiation: Emerging / Developing / Secure
- Formative assessment opportunities throughout

GOOD LUCK TO ALL STUDENTS!